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A systematic mapping and review on machine learning for non-terrestrial networks assisted Internet of Things: Enabling technologies

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ABSTRACT

Non-terrestrial networks (NTNs), comprising unmanned aerial vehicles (UAVs), high-altitude platforms (HAPs), and satellites, are increasingly studied as complements to terrestrial infrastructures for extending connectivity to Internet of Things (IoT) devices and sensor networks. NTN-assisted IoT has been investigated across diverse application scenarios, including environmental monitoring, data collection, emergency response, intelligent transportation, disaster management, remote sensing, and smart city services. To address challenges such as energy constraints, dynamic network conditions, and heterogeneous communication links, a wide range of machine learning (ML) techniques have been explored, including deep learning, reinforcement learning, deep reinforcement learning, and federated learning. These methods have been applied to problems such as energy efficiency optimization, resource allocation, data collection, trajectory planning, task scheduling, and security enhancement. Given the rapid growth of this research area, with hundreds of related publications identified, developing a structured overview remains challenging. This paper addresses this need by presenting a large-scale bibliometric and thematic cluster analysis of ML-based NTN-assisted IoT research with a focus on enabling technologies. The analysis organizes the literature into thematic clusters, examines co-occurrence patterns among enabling technologies and applications, and highlights recurring research directions and emerging topics such as edge intelligence, reconfigurable intelligent surfaces, non-orthogonal multiple access, digital twins, and explainable AI. Finally, the paper discusses current challenges and outlines potential directions for future research, with an emphasis on scalable and deployable NTN-assisted IoT systems.

1. Introduction

Non-terrestrial networks (NTNs), comprising aerial platforms (e.g., UAVs, LAPs, HAPs) and satellite systems across LEO, MEO, and GEO orbits, are increasingly integrated with terrestrial infrastructures to provide connectivity across diverse application scenarios. The mobility, agility, and deployment flexibility of UAVs or drones make them particularly suitable for deployment in disaster-stricken regions, places difficult to reach by humans, or as relays for terrestrial communication. On the other hand, the quasi-stationary positioning of HAPs enables extended coverage in remote and sparsely populated rural areas. Satellites also play a critical role in addressing large-scale coverage gaps and connectivity limitations inherent in terrestrial networks to meet the demands of future 6G Internet of Things (IoT) [1]. Overall, NTNs find applications in remote sensing, the IoT, wireless sensor networks, mobile communications, including 5G and emerging 6G systems, vehicle-to-vehicle communication, satellite broadcasting, and smart cities.

IoT systems consist of distributed smart devices that sense and collect environmental data, which are often transmitted to remote entities

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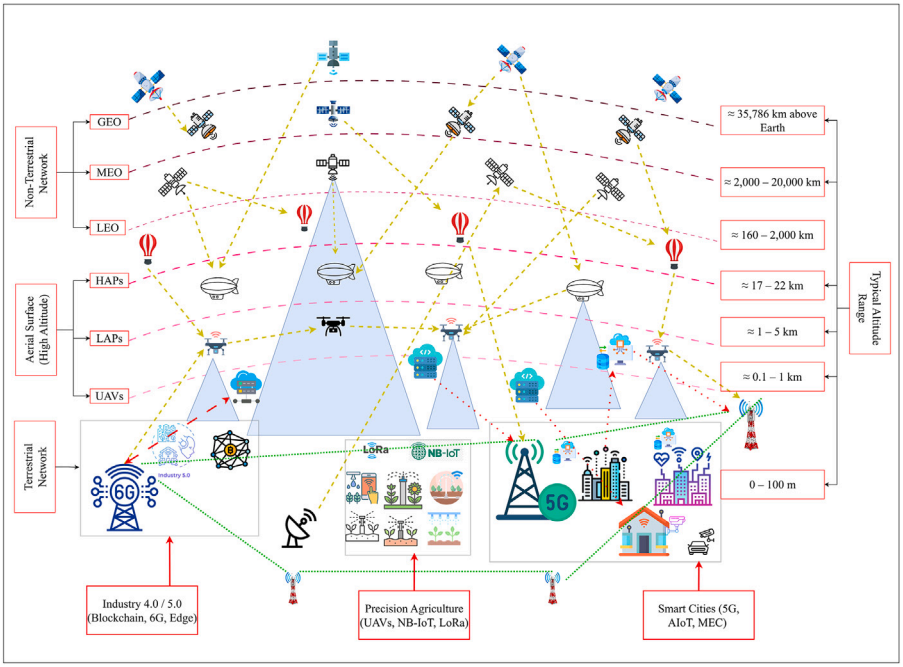

Fig. 1. Hierarchical architecture of integrated terrestrial, aerial, and non-terrestrial networks for 6G-enabled IoT ecosystems.

for processing, analysis, and decision-making across domains such as smart cities, healthcare, environmental monitoring, and security.

With the growing scale, heterogeneity, and dynamic behavior of NTN-assisted IoT deployments, conventional model-based optimization techniques often struggle to cope with uncertainties in network topology, traffic demand, channel conditions, and energy availability. Consequently, machine learning (ML) techniques have been increasingly adopted in NTN-assisted IoT systems to enhance network performance in terms of throughput, energy efficiency, latency, and age of information. Beyond performance optimization, ML algorithms can also be employed to control transmission policies, optimize the trajectories of aerial and satellite platforms, and allocate network resources such as transmit power and bandwidth in adaptive and data-driven manners. AI can also effectively support massive seamless access in space-air-ground integrated networks

Fig. 1 illustrates a hierarchical architecture integrating terrestrial, aerial, and non-terrestrial network segments to support ubiquitous 6G-enabled IoT connectivity. The layered structure emphasizes the complementary roles of terrestrial networks in enabling dense IoT deployments and edge intelligence, aerial platforms such as UAVs, LAPs, and HAPs in providing flexible regional coverage and rapid service provisioning, and satellite systems across LEO, MEO, and GEO orbits in ensuring wide-area and global connectivity. Together, these coupled layers form the architectural basis for scalable, resilient, and low-latency NTN-assisted IoT services envisioned in future wireless ecosystems. At the same time, managing such integrated space–air–ground infrastructures introduces significant challenges related to scalability, energy efficiency, resource coordination, and real-time decision-making, which further motivates the use of data-driven and learning-based approaches.

Despite extensive research on aerial- and satellite-assisted IoT systems and the application of machine learning across several domains, most existing studies tend to examine these topics in relative isolation. As a result, the literature remains fragmented, making it difficult to uncover interdependencies among technologies relevant to NTN-assisted IoT or to identify overarching research themes. This fragmentation further complicates efforts to systematically understand how multiple technologies jointly contribute to NTN-assisted IoT systems, as such relationships often only emerge after reviewing a large and diverse body of work.

Systematic bibliometric analysis offers a structured means to organize large volumes of literature and to reveal both established and emerging research directions. However, such approaches remain relatively underexplored in the context of ML-supported NTN-assisted IoT, particularly from a technology-driven perspective. To address this gap, this paper employs large-scale bibliometric mapping and thematic cluster analysis using VOSviewer, providing an integrated overview of research trends and technological interconnections in ML-based NTN-assisted IoT.

Although bibliometric methods are less common in wireless communication research, they offer valuable capabilities for identifying dominant research themes, and reveal latent relationships among applications, enabling technologies, metrics, and support a more coherent synthesis of the field. Motivated by this, the study pursues three main objectives:

- To systematically map the ML-based NTN-assisted IoT research landscape through a detailed bibliometric keyword co-occurrence analysis, highlighting linkages among technologies.
- To develop a cluster-driven taxonomy of enabling wireless technologies in ML-enabled NTN-assisted IoT research and assess the state of the art across different thematic areas, with particular emphasis on technology-driven clusters.
- To identify underexplored issues, emerging subfields, and promising cross-cutting directions, such as federated learning in NTNs, RIS- and NOMA-enabled edge intelligence, and AI-driven reliability and security for large-scale IoT deployments.

Accordingly, the paper is organized as follows. Section 2 presents related reviews categorized into the key domains of energy efficiency, 5G, and smart cities, while discussing the identified research gaps and motivations. Section 3 details the methodology adopted for this research. Section 4 provides a structural overview and taxonomy of the thematic clusters prioritized in this study, while Section 5 and Section 6 offer a deeper examination of the primary technological intersections and their inter-relationships as identified through VOSviewer clustering. Section 7 briefly evaluates the current state of research across main themes by reviewing representative contributions and real-world deployment scenarios as case studies. Section 8 explores emerging

topics and future research directions. Finally, Section 10 provides the concluding remarks. A comprehensive analysis of all identified themes (beyond core technology-related clusters) would exceed the scope of a single manuscript; consequently, this review focuses on the most influential technological intersections, while remaining clusters are addressed in separate dedicated studies. Given the extensive nature of this review, readers seeking a quick thematic overview should refer to Section 4, while those focused on emerging niches are directed to the directions presented in Section 8.

2. Related reviews

Several review papers have examined specific aspects of NTN-assisted IoT, however, none have offered the comprehensive and integrative perspective advanced in this study, particularly with respect to enabling technologies. In particular, prior surveys tend to emphasize individual technologies (e.g., UAVs, MEC, security, or 5G/6G) or limited combinations thereof, whereas this work adopts a comprehensive bibliometric approach to jointly capture and relate the technologies that shape emerging NTN-assisted IoT deployments and applications. Consequently, existing surveys do not provide a broad systematic mapping of interactions across technologies to understand how they collectively contribute to integrated NTN-assisted IoT systems for diverse applications.

2.1. Energy efficiency

The authors in [2] survey the role of edge computing and AI in UAV-assisted IoT applications within cellular networks. The review addresses UAV technical aspects such as energy efficiency, autonomous navigation, and communication. Applications include agriculture, infrastructure inspection, and delivery systems. The study also discusses challenges in deploying edge AI for UAV-assisted IoT and proposes future research directions.

IoT enables a wide range of applications by facilitating wireless communication among devices that collect environmental data. The goal is to derive actionable insights that improve quality of life and create new revenue streams. A major challenge, however, lies in ensuring large-scale, reliable, and real-time connectivity. The authors [3] present a comprehensive survey of current and emerging communication solutions for IoT, particularly within cellular wide-area networks and NTNs. They highlight advancements in wireless technologies that support IoT in 5G and beyond, focusing on energy efficiency, reliability, low latency, and scalability. Proposed solutions include grant-free access, short packet communication, NOMA, and on-device intelligence. The paper also outlines a vision for future network paradigms in the 2030s, emphasizing AI and NTN integration. Furthermore, it examines the potential of deep learning (DL) and federated learning (FL) to enhance IoT efficiency and security, identifying future research directions for B5G IoT.

The authors in [4] review the role of ML in disaster and pandemic management. The paper discusses the integration of IoT, UAVs, 5G, and satellite-based systems for disaster response. It highlights how ML algorithms have been applied to disaster prediction, disease diagnosis, pandemic spread monitoring, and post-disaster recovery. Related challenges and opportunities for future research are also discussed. The work in [5] provides a systematic review of surveillance drones in smart cities, focusing on application areas, proposed models, and drone characteristics. The review identifies seven main research domains, one of which is disaster management. It also explores how UAVs, integrated with machine learning, IoT, and cloud computing, can deliver efficient and sustainable solutions for urban resilience.

The authors in [6] examines strategies for addressing energy consumption challenges at multiple levels, including hardware-friendly algorithms, mixed-signal circuits, and advanced technologies.

2.2. 5G

The authors in [7] provide a comprehensive literature review on Deep Reinforcement Learning (DRL) applications in communication and networking. This includes autonomous and decentralized modern networks such as IoT and UAVs. The motivation for considering reinforcement learning is the importance of network entities making local decisions to maximize performance, especially given the uncertainty of the network environment. Reinforcement learning (RL) enables these entities to acquire an optimal policy for making decisions or taking actions, in smaller state-action spaces. However, when the network becomes complex and large-scale, the state–action space significantly increases, making it difficult to deploy RL to obtain an optimal policy within a reasonable time. To address these challenges, the combination of RL with Deep Learning, i.e., DRL, has been developed. The authors are motivated to provide a review of DRL approaches for addressing emerging issues in wireless communication and networking, such as dynamic network access, data rate control, wireless caching, data offloading, networking, and connectivity preservation. These areas are fundamental to achieving next-generation networks such as 5G and beyond. In addition, the authors highlight the application of DRL for traffic routing, resource slicing, and data collection. Finally, they discuss key challenges, open issues, and future research directions related to the application of DRL in this domain.

This paper [8] presents the evolution of the NTNs in 5G networks and its role in the development of the 6G ecosystem. The authors discussed key aspects of NTN integration into Terrestrial Networks (TNs) and their synergies, including new services and use cases, various architectures, technological enablers, and relevant higher-layer considerations. They examine challenges arising from the technical constraints and the new approaches being adopted to develop efficient integrated ground-air-space (GAS) networks. It also includes major progress and outcomes from academic research and industry, including field trials, and prototyping toward 6G networks.

RSMA has emerged as a promising multiple access framework for beyond-5G and 6G networks, addressing key limitations of OMA, NOMA, and SDMA. By enabling flexible message splitting and interference management, RSMA significantly improves spectral and energy efficiency while providing a unified framework that generalizes existing multiple access techniques. It supports a wide range of applications, including IoT, UAVs, satellite communications, and massive MIMO, making it a strong candidate for future wireless networks [9]. While RSMA optimizes the transmitted signals to better manage multiuser interference, Reconfigurable Intelligent Surfaces (RIS) complement these efforts by passively reconfiguring the wireless propagation environment. Specifically, RIS can improve UAV communication by enhancing efficiency and reducing energy consumption. RIS can improve spectral efficiency and reduce energy consumption in UAV communication. The study in [10] surveys RIS-assisted UAV systems, with emphasis on optimization strategies, deep reinforcement learning, security considerations, and IoT applications. The survey also identifies several open research challenges, particularly in RIS phase-shift design, channel modeling, and federated learning, that must be addressed to enable robust and scalable RIS-assisted UAV-IoT deployments.

Considering the widespread rollout of 5G in many countries, with the number of subscribers rapidly increasing, the authors in [11] focus on 6G, providing a comprehensive survey of its drivers, use cases, scenarios, requirements, KPIs, architecture, and enabling technologies. They first address whether 6G is necessary, evaluating its potential and possible applications in light of growing mobile traffic. Next, they compare 6G requirements with those of 5G in terms of KPIs. The authors also summarize ongoing 6G research from leading institutions and countries, offering a tentative roadmap for definition, specification, standardization, and regulation. Finally, they identify potential enabling technologies, outlining their benefits, challenges, and open

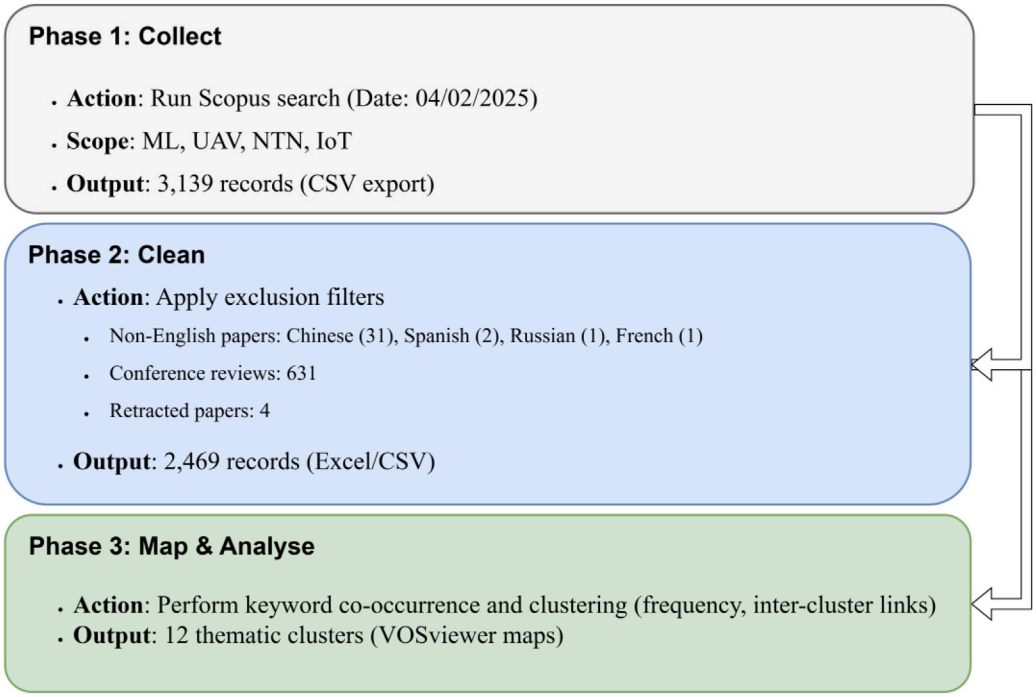

Fig. 2. Data gathering, cleaning, and cluster analysis process and strategies.

research issues, and conclude with a discussion on "what 6G may look like".

IoT encompasses devices that wirelessly communicate data gathered from the environment across a wide range of emerging applications. This capability supports solution development, improves quality of life, and generates revenue. The authors in [3] provide an extensive survey of current and emerging solutions for IoT applications across cellular, wide-area, and NTNs. They present enhancements in wireless communication for 5G and beyond, including unlicensed spectrum usage. The survey also reviews standards and solutions that ensure energy efficiency, low latency, reliability, and scalability, such as grant-free access, short-packet channel coding, NOMA, and on-device intelligence. Moreover, the authors discuss paradigm shifts expected in future IoT, highlighting the integration of AI and NTNs. They specifically emphasize the roles of deep learning and federated learning in improving IoT efficiency and security, along with the challenges and future research directions.

Technologies such as AI and IoT enable smart farming by supporting data collection, transmission, storage, analysis, and decision-making. IoT is particularly central to applications like irrigation systems that monitor water levels, climate conditions, and efficiency. The authors in [12] highlight UAV and robotic applications for seeding, harvesting, irrigation, weed detection, pest spraying, and livestock monitoring. They emphasize the importance of deploying 5G to support such systems, as it leads to high-speed data transfer and large-scale device connectivity. Additionally, the work discusses real-time soil analysis and mapping for informed agricultural decision-making.

The authors in [9] present a tutorial on beyond-5G (B5G) and 6G technologies, covering SDMA, multi-user communication, and massive MIMO as used in 4G and 5G. They examine the link between multi-user communication and multiple access design in 6G, focusing on interference management. The tutorial introduces the principles of interference control and rate-splitting, outlining the RSMA (Rate-Splitting Multiple Access) framework for downlink, uplink, and multi-cell scenarios. Compared to orthogonal multiple access (OMA), NOMA, and space-division multiple access (SDMA), RSMA offers greater efficiency, flexibility, universality, robustness, and reliability. The paper also reviews RSMA applications, addresses myths and frequently asked questions, and proposes directions for future research.

Future internet architectures are expected to be heterogeneous and decentralized, requiring numerous network entities to make local decisions under dynamic and uncertain conditions. Single-agent DRL has been applied to optimize policies through environment interaction, but is fundamentally ill-suited for modeling cooperation or competition among network entities, resulting in the non-stationarity problem. By contrast, multi-agent reinforcement learning (MARL) allows each entity to learn its policy by considering both the environment and the strategies of other agents, improving efficiency. The authors in [13] review MARL applications in communication networks, addressing problems such as transmit power control, content caching, packet routing, trajectory design, computation offloading, and network security. They also discuss challenges and open issues for the deployment of multi-agent DRL in future networks.

The work in [3] provides a comprehensive review on the recent advances of IoT applications in wide-area, cellular and non-terrestrial communications. Specifically, the wireless access solutions such as grant-free and non-orthogonal multiple access were discussed. The review focuses on the solutions that are aligned with the 5G and B5G performance indicators such as energy efficiency and low latency. Further, the AI technologies that help to improve the efficiency and security of the IoT systems were presented as well.

This survey [14] examines physical layer security (PLS) as a solution for enhancing security in 5G networks amid increasing user traffic. It reviews existing PLS schemes using technologies like MIMO, mmWave, and NOMA, and their applications in HetNets, IoT, and more. The survey includes a discussion on AI and machine learning techniques, particularly reinforcement learning and deep learning, for improving PLS. It also addresses challenges in implementing PLS and suggests future research directions.

2.3. Smart cities

The work in [15] presents a systematic review of the role of ML and Deep Learning (DL) in managing smart city applications such as smart power systems, city surveillance, smart buildings, and healthcare

monitoring. The state-of-the-art techniques are categorized according to emerging technologies, including edge computing, cloud computing, fog computing, the Internet of Drones (IoD), and the Internet of Vehicles (IoV).

Smart cities provide numerous benefits to urbanization, including energy management and improved living standards through the adoption of ICT. ICT plays a critical role in policy design, decision-making, and the delivery of productive services. The authors [16] review the role of ML, AI, and DRL in the evolution of smart cities, highlighting how these technologies help design optimal policies for complex urban challenges. Applications reviewed include intelligent transportation systems (ITS), cybersecurity, and energy-efficient smart grids. The use of UAVs to enable optimal services in 6G, B5G, and healthcare contexts is also discussed. Finally, research challenges and future directions are identified, underlining where these technologies can contribute to realizing the vision of smart cities.

The growing demand for compute-intensive applications, driven by IoT and 5G, poses challenges for resource-constrained user devices. Mobile Edge Computing (MEC) has emerged as a key technology to optimize resources by hosting compute-intensive applications at the edge. MEC can process large volumes of data before they reach the cloud, extend cloud computing capabilities to the Radio Access Network (RAN), and provide context-aware services using RAN information. This enables applications with strict real-time requirements, such as augmented reality, robotics, autonomous vehicles, and immersive media. The authors [17] provide a comprehensive overview of MEC technology, its applications, testbeds, and experimental evaluations, as well as open-source efforts. They also discuss lessons learned and outline future research directions for MEC integration with 5G and B5G.

The authors in [16] further review AI techniques applied in smart city contexts, focusing on ML and DRL for optimizing challenges across UAV services in 5G and B5G networks. Similarly, the authors in [18] review caching techniques and enabling technologies for fog computing in IoT-based smart cities. Their study emphasizes the role of UAVs and ML in caching for fog-based IoT systems, along with the associated challenges.

Nikitas et al. [19] define the conceptual intersections of AI, transportation, and smart cities, evaluating their collective impact on urban development. The study examines key smart mobility applications, including Mobility-as-a-Service (MaaS), Connected and Autonomous Vehicles (CAVs), and both personal and unmanned aerial vehicles (PAVs and UAVs). Additionally, the authors resolve terminological ambiguities within smart mobility to better situate these technologies within the framework of urban innovation.

The study in [20] systematically reviews disaster and rescue management in smart cities through UAV path planning and 5G networks. Path planning algorithms are classified into four categories: model-based, conventional, learning-based, and cell-based approaches. Key challenges identified include path optimality, completeness, and efficiency. Broader UAV communication challenges are also discussed, including resource allocation, deployment, and data security. The study concludes by proposing a holistic smart city management system for disaster response and governance enhancement.

The authors in [2] survey the integration of edge computing and AI in UAV-assisted IoT applications within cellular networks. They provided a review on UAV-related aspects such as energy efficiency, autonomous navigation, and communication, and discusses IoT applications in agriculture, infrastructure inspection, and delivery systems. Challenges in deploying edge AI for UAV-assisted IoT are presented, along with future research opportunities.

The United Nations has raised concerns about global warming and its impact on natural resources, particularly glaciers. In response, advanced technologies have been adopted to monitor and predict climate change impacts on these critical reservoirs. Thus, the study in [21] explores the role of emerging technologies, including remote sensing, IoT, AI, UAVs, and robotics, in glacier monitoring and digitalization. By integrating AI, ML, and DL with spatio-temporal data, the study highlights a superior capacity for capturing complex, non-linear patterns in high-altitude environments. Key recommendations include developing intelligent ecosystems for glacier environments, deploying low-cost IoT devices with energy-efficient algorithms, enabling ML-based edge computing, designing glacier-resistant rescue robots, and introducing wearable IoT-based safety devices. These innovations aim to improve predictive capabilities, safety, and global climate resilience.

Pham et al. [17] highlighted the role of mobile edge computing (MEC) in supporting autonomous systems and real-time services in smart cities. Similarly the authors in [22] reviews the integration of ML into structural health monitoring (SHM), highlighting a shift from traditional damage detection to intelligent solutions. It discusses ML pipelines, IoT integration, and advancements in mobile devices, UAVs, augmented/virtual reality (AR/VR), and digital twins. Future challenges and research opportunities for SHM-ML integration are identified.

The authors in [23] investigate the adoption of IoT, UAVs, and the Internet of Underground Things (IoUT) in Agriculture 4.0. They provided a review on the applications of these technologies alongside big data analytics and ML in farm management, presenting use cases and highlighting gaps for future research.

The authors in [24] explore smart farming through information and communication technologies (ICTs) such as UAVs, UGVs, ML, and big data. These technologies are shown to increase productivity, reduce costs, and improve product quality. Similarly, they highlighted key research initiatives and projects across Europe.

The authors in [25] discuss AI's transformative role in 6G communication networks, with emphasis on advancing the intelligent Internet of Things (IoT), augmented human intelligence, and the transition from connected things to connected intelligence. Overall, the work focuses on applying AI to meet emerging human-centric and technological demands.

2.4. Literature gap and motivation

Although the existing literature spans diverse and important themes, ranging from precision agriculture and UAV-assisted mobile edge computing to Age of Information, energy efficiency, 5G/6G technologies, smart cities, big data, and security, these works often examine topics in isolation, concentrating on specific applications, technologies, or domains. While valuable, such fragmented perspectives make it difficult to obtain a holistic understanding of the interconnections, thematic overlaps, and emerging trends that define the broader research landscape. In particular, when enabling technologies such as RIS, NOMA, RSMA, MEC, federated learning, digital twins, and edge intelligence are examined independently, existing surveys may not fully reflect how these technologies are frequently discussed together within NTN-assisted IoT research, as indicated by their repeated co-occurrence in the literature.

Furthermore, most current surveys and reviews rely primarily on narrative synthesis or conventional categorization, with limited or no use of systematic bibliometric mapping to reveal the structural organization of the field based on statistically grounded clustering tools such as VOSviewer. The limitation, therefore, is not the availability of bibliometric tools themselves, but the lack of large-scale, evidence-driven synthesis that explicitly links enabling technologies and application contexts into an interpretable research structure for NTN-assisted IoT.

This paper addresses this gap by conducting a comprehensive bibliometric review of 2469 publications using VOSviewer-based cluster analysis, complemented by an in-depth thematic discussion of the identified clusters. The role of VOSviewer in this study is to enable clustering based on keyword co-occurrence, while the primary contribution lies in interpreting these clusters to uncover recurring technological associations and to extract application-relevant insights for NTN-assisted

Table 1
Overview of identified thematic clusters and representative keywords.

Cluster	Thematic focus	Representative keywords
1	Smart and Sustainable Agriculture Systems	precision agriculture, smart farming, digital agriculture, crop monitoring, remote sensing, environmental monitoring, predictive analytics, sustainability, food security
2	UAV-Enabled Edge Computing and Resource Optimization	mobile edge computing, computation offloading, task offloading, multi-UAV, trajectory design, resource allocation, NOMA, RIS, wireless power transfer
3	Vision, Perception, and Digital Twin Technologies	computer vision, convolutional neural network, image processing, digital twin, data fusion, satellite imagery, predictive maintenance, TinyML
4	Autonomous Aerial Systems and Cyber-Physical Integration	autonomous navigation, UAV systems, obstacle avoidance, cyber–physical systems, robotics, satellite IoT, quality of service, resource management
5	Trajectory-Aware Learning and AoI Optimization	age of information, deep reinforcement learning, trajectory optimization, scheduling, energy efficiency, RIS, satellite networks, smart grid
6	Energy Efficiency and Integrated Enabling Technologies	5G networks, energy efficiency, NOMA, RIS, mobile-edge computing, cognitive radio, disaster management, task scheduling
7	Next-Generation Wireless Communication and IoT Applications	5G, 6G, and Beyond-5G (B5G), network slicing, NB-IoT, LoRa, random access, reliability, smart cities
8	IoT Applications, Cloud–Fog Computing, and Smart Services	cloud computing, fog computing, data analytics, healthcare, surveillance, automation, forest fire detection, augmented reality
9	Security, Privacy, and Internet of Drones	cybersecurity, intrusion detection, IoT security, authentication, privacy, energy harvesting, Internet of Drones
10	Localization, Privacy, and Smart City Analytics	localization, GPS, RSSI, LoRaWAN, smart city, differential privacy, edge computing, GIS
11	UAV Communication, QoS, and Blockchain Integration	5G network, blockchain, Internet of Drones, channel allocation, power control, quality of service, LiDAR
12	Embedded and Energy-Constrained IoT Systems	embedded devices, real-time monitoring, forecasting, energy, low-power sensing

Table 2
Top selected keyword connections across wireless technology-related clusters.

SN	Keyword	Internal connections	External connections
	Energy Efficiency and 5G-Integrated Technologies (Cluster 6)		
1	**Cognitive Radio**	Energy efficiency, NOMA, 5G	5G, B5G, NOMA
2	**Energy Efficiency**	5G, cognitive radio, disaster management, RIS, NOMA	AoI
3	**Non-Orthogonal Multiple Access**	Blockchain, IRS, energy efficiency, disaster management	–
4	**Intelligent Reflecting Surface**	Blockchain, energy efficiency, fairness	AoI, autonomous navigation
	Next-Generation Wireless Communication and IoT Applications (Cluster 7)		
1	**5G**	Wireless communication, MDP, NOMA, B5G, network slicing, random access, NB-IoT, reliability, 6G, SDN, LoRa	Cognitive radio, digital twin, SWIPT, MEC, unsupervised learning, PLS, smart grid, satellite network, communication, security, cybersecurity, blockchain, latency, cyber security, earth observation, cloud computing, application, predictive analytics, sustainability, smart agriculture, edge computing, localization
2	**6G**	LoRa, SDN, reliability, random access, network slicing, NOMA, 5G, wireless communication	RSSI, localization, Industry 5.0, MEC, AoI, task offloading, computation offloading, unsupervised learning, blockchain, cybersecurity, monitoring, latency, edge computing, digital twin, cloud computing, IRS, NOMA, IRS, trajectory design
3	**Air Pollution**	Decision tree, LoRa, satellite remote sensing	Agriculture
4	**B5G**	5G, 6G, network slicing, NOMA, NB-IoT	MEC, cognitive radio, blockchain, Industry 4.0
5	**LoRa**	Air pollution, 5G, 6G	CNN, blockchain
6	**NB-IoT**	6G, B5G, 5G	Edge computing, computer vision, blockchain, 5G network, energy efficiency, resource allocation
7	**NOMA**	5G, B5G, 6G	Cognitive radio, HAP, energy harvesting, security, blockchain, convolutional neural network
8	**Random Access**	NB-IoT, 5G, 6G, B5G	Blockchain, CNN

IoT research. By doing so, this paper attempts to map the intellectual structure of the field and critically examine mature thematic themes, and emerging directions of ML-driven NTN-assisted IoT research, with particular emphasis on enabling wireless technologies.

In line with the technology-oriented direction of this study, the subsequent thematic discussion places greater emphasis on the clusters that most directly relate to real-world NTN-assisted IoT deployments and next-generation wireless services, while the remaining clusters are discussed to provide essential contextual and technological background. This approach offers researchers and practitioners a consolidated reference that supports a clearer understanding of existing research patterns and facilitates the identification of coherent directions for future investigation.

3. Methodology

Building on the prior experience in bibliometric studies and trend analysis, this paper adopts a systematic bibliometric methodology to provide a comprehensive overview of ML-based NTN-assisted IoT research with a deep dive into technology-related clusters. The study collects data from the Scopus database and employs VOSviewer to map and analyze the literature dataset, conducting a bibliometric assessment of research on Machine Learning for Non-Terrestrial Networks. The main goals are to examine research trends, and reveal dominant topics, technologies and their relationships within the literature based on a deeper look into technology-oriented clusters. The methodology, illustrated in Fig. 2, details the process of data collection, cleaning, and visualization, ensuring a rigorous analysis, in accordance with a systematic approach. VOSviewer is employed for bibliometric mapping and clustering; the contribution of this work lies in interpreting the resulting clusters based on keyword co-occurrence, with emphasis on enabling technologies and application-driven NTN-assisted IoT deployments.

3.1. Data collection

The data for this study was gathered from the Scopus database on 04/02/2025. Scopus was selected due to its broad coverage of peer-reviewed journal articles and conference proceedings in engineering, computer science, and related disciplines relevant to NTN-assisted IoT research. The search terms include (“machine learning” OR “supervised learning” OR “unsupervised learning” OR “semi-supervised learning” OR “self-supervised learning” OR “deep learning” OR “neural networks” OR “deep reinforcement learning” OR “drl” OR “Reinforcement learning” OR “federated learning” OR “Edge AI” OR “Edge Learning”) AND (“UAV” OR “Unmanned Aerial Vehicles” OR “drone” OR “Aerial Platforms” OR “NTN” OR “Non-Terrestrial Networks” OR “HAPS” OR “satellite” OR “LEO” OR “MEO” OR “GEO”) AND (“IoT” OR “Internet of Things” OR “sensor networks” OR “WSN”). The search yielded 3139 results which were filtered by excluding non-English content (Chinese-31, Spanish-2, Russian-1, French-1), conference reviews (631), and retracted papers (4). This resulted in a total of 2469 entries exported to CSV format. Furthermore, to enhance thematic clarity and reduce noise, redundant keyword variants and specific low-frequency terms, including technical abbreviations (e.g., DRL, CNN, FL, WSN, and IIoT) and synonymous terms (e.g., drones, satellite communications, HAPS, and sensor networks), were filtered out of the final co-occurrence mapping to prioritize primary technological intersections. Fig. 2 summarizes the research process. The dataset was constructed from a single, documented extraction date and reflects the state of NTN-assisted IoT research at the time of analysis based on keyword co-occurrence patterns.

3.2. Co-occurrence analysis

First, we upload the collected data from Scopus into VOSviewer to generate a mapping of frequently occurring terms within the ML-based NTN-assisted IoT bibliometric dataset. Specifically, a keyword co-occurrence analysis is performed to identify dominant terms and their co-appearance relationships, which are algorithmically grouped into thematic clusters representing major research directions. Twelve thematic clusters are identified (see Table 1), capturing the diverse aspects of ML-based NTN-assisted IoT research, excluding purely methodological keywords and generic architectural components. Details of the internal and external keyword connections are provided in Table 2.

Rather than treating all clusters with equal analytical depth, the subsequent sections use the cluster structure primarily to provide context, while emphasizing technological-focused clusters that are linked to NTN-assisted IoT deployments.

4. Cluster analysis and findings

This section presents the thematic structure of the ML-based NTN-assisted IoT research landscape as revealed by the VOSviewer keyword co-occurrence analysis. In total, twelve distinct clusters were identified, reflecting the diversity of research directions spanning enabling wireless technologies, learning paradigms, system architectures, and application domains. Together, these clusters provide a comprehensive view of how different research themes intersect and evolve within the broader NTN-assisted IoT ecosystem. To facilitate a clear and coherent progression from bibliometric evidence to focused technical discussion, we first provide a high-level overview of all identified clusters before narrowing the analytical scope to selected themes that warrant deeper examination.

4.1. Overview of the identified thematic clusters

The VOSviewer-based keyword co-occurrence analysis resulted in twelve distinct thematic clusters, reflecting the breadth of research topics in ML-based NTN-assisted IoT. These clusters collectively capture a wide spectrum of enabling technologies, system architectures, and application domains, ranging from physical-layer communication techniques and spectrum access to edge intelligence, security, smart cities, agriculture, and disaster management. Table 1 provides a concise overview of all twelve thematic clusters using representative keywords identified through the VOSviewer analysis, establishing the broader research context before the focused discussions that follow. To provide a holistic perspective, we first summarize the thematic scope of all identified clusters at a high level.

4.2. Cluster extraction and connectivity-guided interpretation

VOSviewer is commonly used in systematic reviews for bibliometric mapping and cluster identification. In this study, it is employed as a standard tool to construct keyword co-occurrence networks within the domain of wireless communication technologies for NTN-assisted IoT. Rather than relying solely on visual inspection of bibliometric maps, the analysis is complemented by a connectivity-guided interpretation of the resulting clusters to improve consistency and transparency in thematic labeling.

For each identified cluster, both keyword occurrence frequency (as reported by VOSviewer) and intra-cluster co-occurrence patterns are examined. Keywords exhibiting strong connectivity within a cluster are used as reference terms for characterizing its dominant technical focus. This approach reduces reliance on subjective visual prominence and helps ensure that cluster descriptions are grounded in the underlying co-occurrence structure of the literature.

Such connectivity-aware interpretation is particularly relevant for wireless communication research, where enabling technologies, such

Table 3
Selected keyword statistics for energy efficiency and integrated enabling technologies cluster.

Keyword	# Conn.	Occ.
Energy Efficiency (EE)	5	5
Non-Orthogonal Multiple Access	4	8
Cognitive Radio (CR)	3	6
Intelligent Reflecting Surface (RIS)	3	6

as spectrum access mechanisms, energy-efficient design, reliability, and edge intelligence, are often studied jointly and exhibit strong interdependencies. By reflecting these relationships explicitly, the analysis provides a more coherent view of research trends in ML-based NTN-assisted IoT.

4.3. Analytical focus and thematic emphasis

As summarized in Table 1, all twelve clusters contribute to the broader landscape of ML-based NTN-assisted IoT research. However, the clusters whose dominant themes emphasize energy efficiency, spectrum access and reliability, and next-generation wireless integration (clusters six and seven) are most directly aligned with wireless communication systems and their enabling technologies. Accordingly, these clusters are examined in greater depth in the remainder of this paper.

The remaining clusters are briefly outlined to provide contextual completeness, while the primary discussion focuses on enabling wireless communication technologies relevant to deployable NTN-assisted IoT systems.

Each subsequent subsection highlights the representative keyword composition of the selected clusters, their principal research directions, and relevant cross-theme connections. A detailed, cluster-by-cluster analysis of all identified themes would exceed the scope of a single manuscript. Consequently, this review focuses on the most influential technological intersections within two primary clusters, while the remaining clusters are addressed in separate dedicated studies.

4.4. Energy efficiency and integrated technologies

This subsection focuses on energy efficiency and integrated enabling technologies, which emerge as a dominant research theme in ML-based NTN-assisted IoT, particularly in 5G-enabled and UAV-assisted systems. We provide a broad overview of this research theme here, establishing a foundation for the comprehensive discussion and interconnection between terms presented in Section 5.

This theme comprises 17 keywords, including *5G networks, blockchain technology, cognitive radio (CR), data analysis, disaster management, energy efficiency (EE), fairness, intelligent reflecting surface (RIS), joint optimization, mobile-edge computing (MEC), non-orthogonal multiple access (NOMA), nonorthogonal multiple access, trajectory optimization, trajectory planning, task scheduling, and wireless power transfer (WPT)*. The most strongly connected keywords are summarized in Table 3, which form the analytical emphasis of this subsection.

The cluster centers on methods for improving energy efficiency in mission-critical NTN-assisted IoT, particularly UAV-based systems. Key enabling technologies include NOMA, RIS, MEC, CR, and WPT, each supporting energy-efficient communication and resource allocation. Energy efficiency is closely linked to performance metrics such as fairness and trajectory optimization, which indicates the need for balanced service provisioning in multi-objective optimization scenarios. From a system-level perspective, this cluster emphasizes the importance of improving energy efficiency through the joint consideration of communication efficiency, enabled by NOMA, cognitive radio, and RIS-assisted transmission in mobile NTN-IoT systems (e.g., UAV-mounted RIS), under stringent energy constraints. In this context, fairness-aware energy allocation has been shown to balance energy consumption across distributed IoT devices operating under heterogeneous traffic demands, residual energy levels, and channel conditions [26]. These system-level considerations are particularly critical in reliability- and security-sensitive NTN scenarios, such as satellite- and UAV-assisted IoT deployments for disaster response and other time-critical services.

Table 4
Keyword statistics for next-generation wireless communication technologies' cluster.

Keyword	# Conn. Keywords	Occ.
5G	11	34
6G	8	37
B5G	5	9
Random Access	4	8
NOMA	3	13
LoRa	3	10
Network Slicing	3	6
NB-IoT	3	6
Air Pollution	3	6

Applications within this cluster focus on disaster management, secure communication using blockchain, UAV-enabled services, and sustainability-driven IoT deployments. Blockchain can further support research in this cluster by enabling secure task management and computation offloading. Studies on UAV-enabled cognitive IoT systems show that inherent communication characteristics, such as short-packet transmission in IoT and limited onboard energy availability in UAVs, are practical constraints that directly influence achievable energy efficiency and therefore require careful optimization [27]. In addition, analytical investigations of cognitive and NOMA-enabled air–ground IoT networks indicate that mobility plays a central role in shaping system performance, affecting both UAV-to-UGV communication reliability and spectrum utilization. In such settings, NOMA-based transmission enables efficient data transfer from UAVs to ground vehicles while supporting simultaneous transmissions and extended coverage beyond base-station connectivity through cognitive operation. Furthermore, these studies reveal a strong interplay between mobility, channel outdatedness, and network performance [28]. Taken together, these findings highlight that, in realistic NTN-assisted IoT deployments, robust performance depends on the coordinated design of communication technologies and system parameters under stringent operational constraints.

Beyond this cluster, connections extend to digital twins, smart agriculture, and Industry 4.0, underscoring the broad applicability of energy-efficient design principles across diverse NTN-assisted IoT domains.

4.5. Next-generation wireless communication

This subsection examines next-generation wireless communication technologies that underpin NTN-assisted IoT deployments, with particular emphasis on 5G, beyond-5G (B5G), and emerging 6G-enabled systems. We provide a broad overview of this research theme here, establishing a foundation for the comprehensive discussion and interconnection between terms presented in Section 6.

This theme comprises 17 keywords related to advanced wireless communication, including *5G, 6G, B5G, wireless communication, smart cities, air pollution, satellite remote sensing, LoRa, NB-IoT, network slicing, NOMA, RIS, SDN, reliability, random access, decision tree, and Markov decision process (MDP)*. The most strongly connected keywords are summarized in Table 4, which guide the discussion in this subsection.

The cluster highlights the foundational role of advanced wireless technologies in NTN-assisted IoT. 5G, B5G, and 6G provide the backbone for enhanced connectivity, supported by enabling mechanisms such as NOMA, RIS, SDN, NB-IoT, and LoRa. Reinforcement learning methods, including MDP-based approaches, illustrate the integration

Table 5
Mapping between identified thematic focus areas and their application domains and key technologies (keyword-based).

Thematic focus area	Application domains	Key Technologies/techniques
Energy Efficiency and Integrated Technologies (Cluster 6)	Disaster Management	5G Networks, Blockchain Technology, Cognitive Radio (CR), Data Analysis, Energy Efficiency (EE), Fairness, Intelligent/Reconfigurable Reflecting Surfaces (RIS), Joint Optimization, Mobile-Edge Computing (MEC), Non-Orthogonal Multiple Access (NOMA), Wireless Power Transfer (WPT)
Next-Generation Wireless Communication (Cluster 7)	Smart Cities, Air Pollution, Satellite Remote Sensing	5G, 6G, B5G, Wireless Communication, LoRa, NB-IoT, NOMA, Network Slicing, RIS, SDN, Decision Tree, Markov Decision Process (MDP)

of AI for dynamic spectrum access, decision-making, and resource optimization. Dynamic and efficient resource allocation is essential in SDN-enabled 6G networks to support adaptability and effective utilization of network resources. Reinforcement learning–based scheduling has been shown to improve responsiveness to traffic variations and reduce packet drops under dynamic conditions, although performance remains sensitive to network faults and outages [29]. Similarly, the inherent heterogeneity and time-varying characteristics of space–air–ground integrated network environments complicate network management and optimization. In such settings, the combination of SDN with reinforcement learning enables multi-level control architectures that support coordinated resource allocation and policy optimization across different network domains [30].

Applications within this cluster span smart cities, satellite-enabled remote sensing, and environmental monitoring, with representative use cases such as air pollution detection. Key performance requirements include reliability, low latency, and efficient random access in large-scale IoT deployments. These performance requirements are particularly challenging in NTN-assisted IoT due to the long propagation and access delays and high collision rates associated with LEO satellite links. Deep reinforcement learning–based handover mechanisms have been shown to mitigate these effects by reducing access delay and collision probability while improving handover success rates [31]. Furthermore, LEO satellite and UAV-assisted IoT networks are characterized by frequent topology changes, constrained bandwidth, and time-varying propagation delays, which limit the effectiveness of centralized routing approaches. As a result, adaptive and distributed learning-based solutions, such as multi-agent reinforcement learning, are increasingly adopted to enable context-aware decision-making based on local observations and to achieve stable tradeoffs across key performance metrics [32].

Taken together, the two thematic focus areas summarized in Table 5 illustrate the convergence of energy-efficient system design and advanced wireless communication technologies in NTN-assisted IoT. The energy-efficiency–driven theme emphasizes sustainable and optimized architectures enabled by EE-aware design, NOMA, RIS, MEC, and WPT, while the next-generation communication theme captures the evolution of wireless systems from 5G to B5G and 6G, supported by technologies such as SDN, NB-IoT, and learning-driven access mechanisms. Their combined application scope spans disaster management, smart cities, UAV-enabled services, and environmental monitoring, underscoring their complementary roles in shaping scalable, resilient, and future-ready IoT ecosystems.

5. Energy efficiency and 5G-integrated technologies

This section examines energy efficiency and integrated enabling technologies that play a central role in ML-based NTN-assisted IoT, particularly within 5G-enabled environments. The distribution and connectivity of the associated keywords are illustrated in Fig. 3, highlighting a tightly connected research theme comprising 17 core keywords highlighted in Section 4.4.

These keywords reflect the applications, technologies, and technical requirements that characterize energy-efficient NTN-assisted IoT systems operating over 5G networks. Key enabling technologies include blockchain, cognitive radio, NOMA, RIS, mobile edge computing, and wireless power transfer, which jointly support efficient communication, computation, and power management. Representative application scenarios are primarily related to disaster management, while key performance objectives include energy efficiency and fairness. Joint optimization is often required to address multiple, potentially conflicting objectives, with task scheduling, trajectory optimization, and trajectory planning emerging as core technical requirements.

The central theme of this cluster revolves around integrated technologies and methods for achieving energy efficiency in Non-Terrestrial Network (NTN)-assisted IoT, particularly within UAV-assisted frameworks and those incorporating 5G networks. Energy efficiency (EE) is a critical concern in such systems, especially in mission-critical applications such as disaster management, where reliable and optimized service provisioning is essential. In these scenarios, prolonged operation time of UAVs, satellites and IoT devices, rapid deployment, and resilient connectivity must be achieved under stringent energy constraints while meeting latency requirements, motivating the joint consideration of communication, computation, and mobility control. To provide a broader perspective on the technological landscape, the subsequent subsections detail the links between terms within the VOSviewer 'energy efficiency and 5G' cluster, providing contextual inferences for these relationships.

5.1. Energy efficiency and integrated technologies

Energy efficiency is connected to 5G, cognitive radio (CR), disaster management, reconfigurable intelligent surfaces (RIS), and non-orthogonal multiple access (NOMA) within the same cluster, and to Age of Information (AoI) in a different cluster. These links highlight EE's pivotal role in 5G-based NTN-IoT and cognitive radio-enabled communication systems. Within disaster response scenarios, energy-efficient communication strategies are imperative, emphasizing both responsiveness and sustainability. From a system-design perspective, these connections indicate that energy efficiency could be jointly optimized with timeliness, fairness, and reliability metrics, rather than being treated as a singular objective.

RIS and NOMA are frequently utilized technologies within energy-efficient NTN-assisted IoT architectures. RIS is linked to trajectory optimization and NOMA in this cluster, suggesting its deployment in scenarios requiring precise path planning and advanced channel access mechanisms. In other clusters, trajectory optimization is associated with energy efficiency, data collection, resource allocation, autonomous navigation, energy harvesting, and AoI. Trajectory planning is further linked to fairness in this cluster, and to access control, edge computing, energy efficiency, data collection, and computation offloading across other clusters. However, practical deployment of RIS-assisted wireless systems is highly sensitive to channel state information (CSI) quality. Analytical studies show that imperfect CSI can lead to significant performance degradation, including outage probability floors and loss of diversity gains at high SNR in RIS-aided links [33]. Moreover, RIS–NOMA system designs that rely on instantaneous CSI typically incur high pilot overhead and computational complexity and statistical CSI-based optimization strategies have therefore been explored as a means to reduce pilot overhead while explicitly accounting for practical impairments [34].

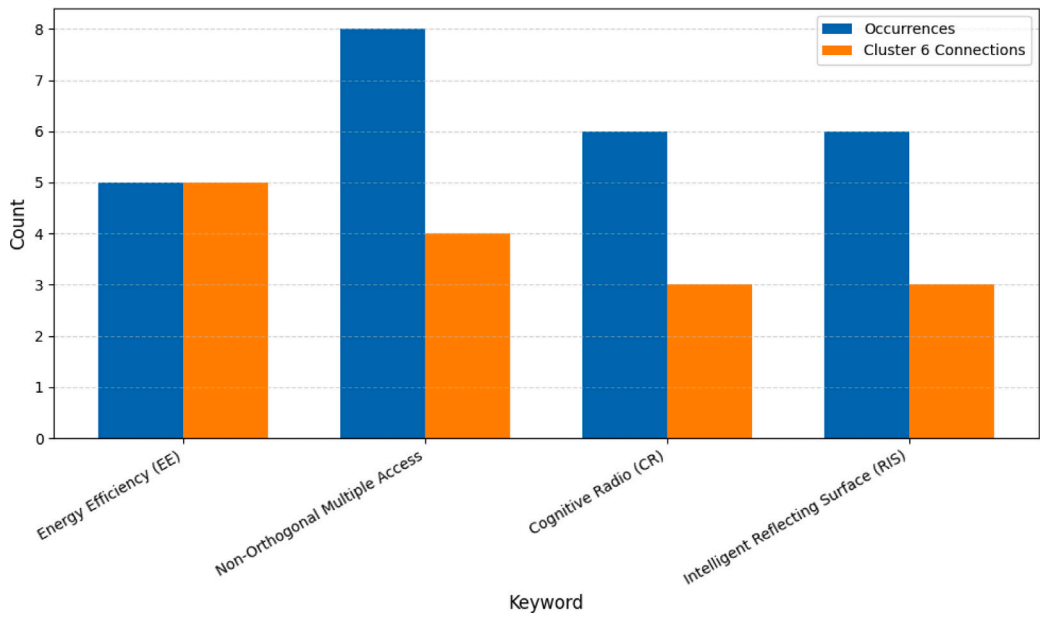

Fig. 3. Keyword distribution and connectivity for energy efficiency and integrated enabling technologies.

5.2. 5G, B5G, and associated technologies

5G is directly connected to cognitive radio in this cluster. Cognitive radio, studied in the context of future wireless systems such as 5G and B5G, supports dynamic spectrum access and enables energy-efficient communication. It is linked with energy efficiency, NOMA, and 5G in this cluster, and with 5G, B5G, and NOMA in other clusters. Additionally, 5G is associated with access control, authentication, blockchain, security, monitoring, cybersecurity, Agriculture 4.0, smart farming, precision agriculture, wireless communication, resource allocation, satellite communication, network slicing, and edge computing in other clusters. These associations reflect the comprehensive set of technical requirements and application areas supported by 5G-enabled IoT architectures, particularly in conjunction with NTN and UAV platforms. These associations point to the increasing complexity of 5G-enabled NTN-assisted IoT systems, where energy efficiency, spectrum management, security, access control, and resource allocation must be carefully considered across multiple network layers across layers, spanning IoT devices, spanning IoT devices, UAVs, and satellites.

5.3. Non-orthogonal multiple access

NOMA is another key enabler of efficient channel access in 5G-based IoT systems. It is linked to blockchain, RIS, energy efficiency, disaster management, and cognitive radio in this cluster, and to clustering, resource allocation, and digital twins in other clusters. Together, these associations highlight the broad applicability of NOMA in 5G-enabled and NTN-assisted IoT architectures, particularly in mission-oriented scenarios where energy efficiency is crucial.

In NTN realizations such as satellite–terrestrial links, analytical studies show that NOMA performance and energy efficiency are highly sensitive to practical impairments, including imperfect channel state information and residual successive interference cancellation, with outage probability, ergodic sum rate, and energy efficiency degrading as residual interference increases [35]. In high-mobility satellite scenarios, time-varying channel conditions further complicate NOMA reception, making interference cancellation a key bottleneck for maintaining reliable performance [36].

5.4. Reconfigurable intelligent surfaces

RIS, which supports intelligent signal reflection and control, is linked to blockchain, energy efficiency, and fairness in this cluster, and to AoI and autonomous navigation elsewhere. This indicates the relevance of performance metrics such as energy efficiency, fairness, and AoI in RIS-enhanced IoT architectures. Fairness, in particular, is studied in conjunction with both RIS and trajectory planning within the cluster. In other clusters, fairness is also connected to power control, MEC, security, data collection, UAV trajectory, and wireless power transfer (WPT), showing its broad relevance due to the importance of ensuring equitable service delivery across diverse IoT settings. These relationships indicate that, beyond communication efficiency, fairness emerges as a recurrent consideration in RIS-enabled NTN-assisted IoT research across diverse application contexts. Based on keyword co-occurrence patterns, fairness is frequently associated with aspects such as resource allocation in MEC, data collection, and power control, particularly in aerial- and satellite-assisted IoT environments.

5.5. Mobile edge computing

MEC is associated with fairness in this cluster, reflecting the goal of fair service provisioning of computing resources for terrestrial IoT users. In other clusters, MEC is linked to game theory, computation offloading, resource allocation, and blockchain. Taken together, these associations suggest that fairness-aware computation offloading and resource allocation are important considerations in MEC-related research for NTN-assisted IoT, particularly in scenarios where a large number of IoT devices compete for limited edge computing resources.

5.6. Data analysis and task scheduling

Data analysis and task scheduling are also linked within this cluster. In other clusters, data analysis is associated with feature selection, classification, precision agriculture, MEC, and data collection. These connections suggest that data collected from IoT devices, often offloaded via UAVs, is analyzed for various applications, including smart farming and health monitoring. Task scheduling, in this context, may involve offloading tasks to MEC infrastructure where they undergo data analysis. Accordingly, learning-based task scheduling is commonly considered in studies addressing task allocation in UAV-assisted IoT systems under dynamic network conditions and energy constraints [37].

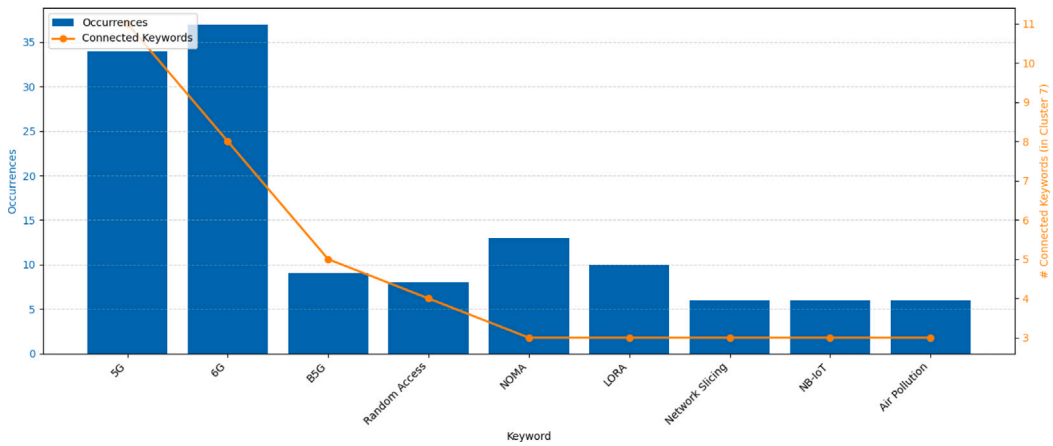

Fig. 4. Keyword distribution and connectivity for next-generation wireless communication technologies.

6. Next-generation wireless communication and IoT applications

This section examines next-generation wireless communication technologies that are fundamental to NTN-assisted IoT systems, with particular emphasis on 5G, beyond-5G (B5G), and emerging 6G-enabled networks. The distribution and connectivity of the associated keywords are illustrated in Fig. 4, revealing a tightly connected research theme comprising 17 core keywords highlighted in Section 4.5.

From this cluster, the role of NTNs in 5G, 6G, and B5G is evident, along with their applications in smart cities and environmental monitoring. In addition, the role of satellite remote sensing in these applications can be inferred from this cluster. Similarly, the IoT communication technologies used in these applications are also observed, such as LoRa and NB-IoT. The role of reinforcement learning is reflected through the presence of the Markov decision process keyword. Furthermore, 5G-, 6G-, and B5G-related enabling technologies are observed, including network slicing, NOMA, RIS, and SDN. In such applications, higher reliability is one of the most important requirements, serving as a key motivation for the transition to 5G and beyond. Another observation from this cluster is the popularity of decision trees in ML-based NTN-assisted IoT research for 5G, as well as the prominence of the random access problem, which relates to the control of device access to the wireless channel for transmission. Taken together, these observations suggest that research on next-generation wireless communication for NTN-assisted IoT frequently brings together advanced access technologies, learning-based control, and reliability considerations across a range of application scenarios.

Cluster 7 highlights the central role of wireless communication technologies, particularly 5G, B5G, and 6G, as key enablers of NTN-assisted IoT. These technologies form the foundation for enhanced communication, reliability, and efficiency in next-generation IoT systems. Overall, this pattern reinforces the close association between learning-oriented decision mechanisms, such as decision trees and Markov decision processes, and reliability and access-related themes in next-generation wireless communication research for NTN-assisted IoT. To provide a broader perspective on the technological landscape, the subsequent subsections detail the links between terms within the VOSviewer 'next generation wireless communication and applications' cluster, providing contextual inferences for these relationships.

6.1. Wireless communication

Wireless communication acts as an overarching theme that binds together 5G and 6G in this cluster. In other clusters, it is connected to game theory, blockchain, edge computing, and digital twins, suggesting the wide range of technologies that integrate with or are enabled by wireless connectivity. This association highlights the role of wireless communication as a foundational layer that supports not only data transmission but also the integration of computation- and service-oriented technologies in NTN-assisted IoT.

6.2. 5G

5G is the most connected keyword in this cluster, linked to wireless communication, Markov decision processes (MDPs), NOMA, B5G, network slicing, random access, NB-IoT, reliability, 6G, software-defined networking (SDN), and LoRa. These associations underscore 5G's pivotal role in supporting massive connectivity and high-reliability communication for IoT. The presence of enabling technologies such as NOMA, SDN, and network slicing within this cluster further reflects the advanced capabilities expected of 5G networks. SDN, while only linked to 5G and 6G within this cluster, supports dynamic network management and is connected to cloud computing, security, and blockchain integration in other clusters. Taken together, these associations highlight that 5G-related research in NTN-assisted IoT frequently brings together access mechanisms, programmable networking, and learning-based decision models to address reliability and connectivity requirements.

6.3. 6G

6G emerges as a prominent technology in this cluster, linked to 5G, LoRa, SDN, reliability, random access, network slicing, NOMA, and wireless communication. Beyond this cluster, 6G connects to RSSI, localization, Industry 5.0, MEC, Age of Information (AoI), task offloading, computation offloading, unsupervised learning, blockchain, cybersecurity, latency, edge computing, digital twins, cloud computing, IRS, and trajectory design. These links reflect growing research interest in 6G for highly reliable, secure, and low-latency IoT systems with advanced offloading and localization capabilities. Notably, the association between RSSI and localization highlights the importance of accurate positioning for autonomous and industrial IoT applications. Overall, these connections indicate that 6G-focused NTN-assisted IoT research frequently links communication and networking mechanisms (e.g., SDN, network slicing, and random access) with sensing- and positioning-related topics (e.g., RSSI and localization), alongside learning- and edge-enabled functions such as unsupervised learning and computation offloading.

6.4. B5G

B5G (Beyond 5G) is linked to 5G, 6G, network slicing, NOMA, and NB-IoT in this cluster, and to MEC, cognitive radio, blockchain, and Industry 4.0 in other clusters. This links are quite similar to 5G links and thus its positioning suggests that B5G serves as a transitional phase that integrates both mature and emerging wireless technologies.

6.5. NOMA

NOMA is linked to 5G, B5G, and 6G in this cluster, and to cognitive radio, high-altitude platforms (HAPs), energy harvesting, security, blockchain, and convolutional neural networks (CNNs) in other clusters. These connections reflect the role of NOMA as an efficient channel access technique and its compatibility with a range of enabling technologies, including aerial platforms such as HAPs.

Taken together, these associations indicate that NOMA-related research in NTN-assisted IoT is often discussed alongside access control, energy-efficient communication, and learning-based techniques. For example, [38] highlights the role of NOMA in supporting massive connectivity and diverse IoT service requirements in 6G networks.

Beyond conventional NOMA designs, related studies have examined high-altitude platforms and UAVs in combination with advanced multiple-access and reconfigurable technologies to improve transmission reliability and resource utilization under heterogeneous channel conditions. In this context, federated DRL frameworks have been investigated to enable distributed optimization of beamforming, RIS reflection coefficients, UAV trajectories, and power allocation, while preserving data privacy through localized training [39]. Such distributed learning approaches can reduce centralized training and signaling overhead compared to fully centralized optimization methods. Recent work has also explored rate-splitting multiple access (RSMA) as an alternative multiple-access strategy in NTN-assisted and space–air–ground integrated networks [40].

6.6. Network slicing

Network slicing, linked to 5G, 6G, and B5G, is vital for providing service-level customization and Quality-of-Service (QoS) guarantees in future networks. In other clusters, it is associated with edge computing, security, and blockchain, reflecting its role in secure and flexible network provisioning. Taken together, these associations suggest that network slicing is frequently discussed alongside edge-enabled service provisioning and security-related themes in NTN-assisted IoT research.

6.7. Random access and reliability

Random access is an important component of 5G communication, contributing to improved reliability, particularly in massive IoT scenarios. It is linked to NB-IoT, 5G, 6G, and B5G within this cluster, and to blockchain and CNNs in other clusters. *Reliability*, another key performance metric, is connected to 5G and 6G here, and to embedded sensors, energy efficiency, edge computing, security, and latency in other clusters, emphasizing the need for robust and responsive communication systems in edge computing and sensing applications. Overall, these associations suggest that reliability is a central concern in research on random access for NTN-assisted IoT, and that security and latency are important considerations in the design and evaluation of random access mechanisms for future wireless communication systems.

Satellite communications are promising due to their wide coverage footprint and ability to support massive access and connectivity of IoT devices. As a result, seamless connectivity and large-scale device access emerge as key requirements for future NTN-assisted IoT systems. Such systems are inherently characterized by dynamic network topologies, limited spectrum resources, time-varying and potentially unstable links, and stringent energy constraints, all of which challenge conventional random access mechanisms. Consequently, recent studies increasingly explore AI-driven approaches to support more adaptive and reliable multiple-access operation in integrated space–air–ground networks, particularly under large-scale and heterogeneous IoT deployments [41].

6.8. NB-IoT and LoRa

NB-IoT and *LoRa* are two low-power wide-area network (LPWAN) technologies linked to 5G in this cluster. NB-IoT is also connected to edge computing, computer vision, blockchain, energy efficiency, and resource allocation in other clusters, highlighting its suitability for energy-constrained IoT deployments. LoRa is linked to air pollution monitoring in this cluster and to CNNs and blockchain elsewhere, emphasizing its role in long-range and low-power environmental monitoring applications. Overall, these associations indicate that NB-IoT and LoRa are commonly positioned as low-power access technologies for wide-area, application-specific scenarios, such as air pollution monitoring, within NTN-assisted communication research.

6.9. Air pollution and remote sensing

Air pollution is connected to satellite remote sensing and decision tree algorithms, suggesting the application of machine learning models for processing remote sensing data in environmental monitoring. *Decision tree* is also linked to smart cities, agriculture, and computer vision in other clusters, indicating its broader applicability across IoT domains.

6.10. Smart cities

The keyword *smart cities* is primarily connected to other clusters through terms such as disaster management, Internet of Drones (IoD), fog computing, blockchain, cloud computing, cybersecurity, and data analytics, highlighting the interdisciplinary nature of smart urban ecosystems.

6.11. Markov decision processes

Finally, *MDPs*, linked to 5G within this cluster, indicate the use of reinforcement learning techniques for decision-making in network management and resource optimization. MDPs are also connected to fog computing and edge intelligence in other clusters, demonstrating their relevance in distributed and adaptive NTN-assisted IoT systems. However, deploying reinforcement learning in complex NTN-assisted IoT scenarios, such as channel learning under mobility, large-scale access control, or dynamic resource allocation, requires carefully balancing performance gains against training overhead, convergence time, and system stability. Algorithm selection must explicitly consider learning efficiency and computational complexity, as high training costs can be prohibitive for energy- and computation-constrained platforms such as UAVs. In such cases, excessive learning overhead may degrade real-time responsiveness and reduce the practical effectiveness of learned policies for target applications [37,42].

In summary, this cluster is characterized by a strong focus on the evolution of wireless technologies (5G, B5G, and 6G), with deep integration of enabling mechanisms such as NOMA, network slicing, SDN, and random access. These technologies are complemented by LPWAN standards such as NB-IoT and LoRa and are directed toward application-driven scenarios including smart cities, environmental monitoring, and industrial IoT. The cluster further reflects emerging interest in reinforcement learning, edge intelligence, and resilient communication infrastructures, positioning Cluster 7 as a core pillar for real-world NTN-assisted IoT deployments.

7. Highlights of research linked to selected keywords

The thematic clusters reveal a landscape of overlapping technical domains, bridging core communication protocols with emerging machine learning applications. This mapping highlights current research trends and shows how technologies like satellite access and edge computing are being integrated. The following sections explore representative works that demonstrate these identified patterns.

Energy efficiency in NTN-assisted IoT applications:. Below, we provide a short overview of representative studies that address energy efficiency and energy consumption in NTN-assisted IoT systems, with particular emphasis on UAV-enabled architectures. The selected works are discussed in relation to the dominant themes identified through the bibliometric cluster analysis, including energy-aware communication, computation offloading, and mobility management.

Arsalan et al. [43] investigate energy-efficient task allocation and seamless service migration in UAV-based IoT networks by leveraging federated learning and digital twin frameworks, highlighting their potential for adaptive coordination under dynamic network conditions. From a communication perspective, Vaezi et al. [3] survey mechanisms such as grant-free access, NOMA, and on-device intelligence, emphasizing their relevance to supporting energy efficiency alongside scalability, reliability, and low-latency requirements in 5G and beyond NTN-enabled IoT systems. Complementing these network-level approaches, Sze et al. [6] discuss energy consumption challenges in machine-learning-enabled systems and outline how hardware-friendly algorithms, mixed-signal circuits, and architectural design choices influence energy efficiency at the device level. At the system optimization level, Liu et al. [44] study a dual-UAV-assisted IoT framework employing NOMA and joint optimization of scheduling, transmission power, and UAV mobility, illustrating how coordinated parameter tuning can reduce UAV energy consumption while maintaining throughput. Taken together, these works suggest that improving energy efficiency in NTN-assisted IoT requires the coordinated consideration of networking technologies, learning-based control, and hardware-aware design–directly reinforcing the thematic patterns identified in our earlier cluster analysis.

Next-generation wireless communication for NTN-assisted IoT:. Representative studies aligned with next-generation wireless communication in NTN-assisted IoT are discussed below, with emphasis on the evolution toward 5G and 6G, learning-enabled optimization, and emerging architectural enablers.

Sharma et al. [45] survey key technologies for 6G networks, including AI/ML-driven optimization, dynamic spectrum management, and advanced air interface design. This work highlights the growing importance of learning-based optimization for adaptive next-generation wireless systems, particularly with respect to efficient spectrum allocation, ultra dense networking, and security. Other studies further emphasize the role of next-generation wireless technologies as key enablers of NTN-assisted IoT systems. Jaiswal et al. [46] investigate computation offloading in 6G-enabled LEO satellite–UAV IoT architectures, highlighting the relevance of integrating space–air platforms with intelligent resource management mechanisms. Complementing this perspective, Al-Hilo et al. [47] examine RIS-assisted UAV architectures for timely data collection, with a focus on reliability and latency considerations in UAV-enabled IoT networks. Chen et al. [48] extend this line of work by applying deep reinforcement learning to optimize energy efficiency in active flying RIS-assisted MEC networks, illustrating how learning-based control can be employed to jointly consider mobility, computation, and communication factors. At a broader architectural level, Azari et al. [49] provide a comprehensive review of non-terrestrial networks and their integration with terrestrial systems, underscoring the importance of ground–air–space architectures in the evolution from 5G toward 6G. Finally, Vaezi et al. [3] contextualize these developments within cellular and wide-area IoT, highlighting how mechanisms such as grant-free access, NOMA, and AI-driven optimization contribute to scalability, reliability, and energy efficiency in NTN-assisted deployments. Collectively, these works point to a convergence of 6G evolution, intelligent networking, and NTN architectures in supporting reliable and scalable IoT systems.

While the above studies primarily focus on associated technologies and AI-based solutions through system models, simulations, or surveys, it is equally important to examine works that demonstrate their application in real operational environments. We therefore conclude this section by highlighting representative real-world deployment scenarios that support and contextualize the findings of the analysis.

7.1. Real-world deployment scenarios and case studies

This subsection highlights representative real-world deployment scenarios drawn from recent literature. Rather than experimental evaluations based on simulations, these case studies illustrate how some of the technologies and research clusters identified in this paper are considered in operational environments. The aim is to further proove the bibliometric findings are relevant in real-world use cases.

7.1.1. Satellite-assisted smart agriculture

NB-IoT integration into satellite communication systems has the potential to extend IoT services beyond the coverage limits of terrestrial infrastructures. In [50], the authors adopt a service-oriented design methodology supported by both link-level and system-level investigations to jointly optimize physical-layer transmission parameters, satellite constellation design, and protocol architecture. To demonstrate a real smart agriculture service in Europe, the proposed solution employs a constellation of 24 LEO satellites arranged across eight orbital planes at an altitude of 500 km. Each satellite hosts the complete protocol stack onboard, enabling terminals to transmit small application-layer payloads (on the order of tens of bytes) without relying on a feeder link. As a result, end-to-end communication latencies ranging from 16 to 75 min are achieved, depending on the number of served terminals and transmission settings. Furthermore, the use of early data transmission schemes reduces latency by up to 40%, illustrating the feasibility of satellite-enabled NB-IoT as a proof-of-concept solution that can significantly shorten time-to-market compared with existing approaches.

7.1.2. Scalable rural IoT connectivity using LoRa

Providing reliable and scalable wireless connectivity in rural environments is essential for many IoT applications, yet it remains challenging due to both economic and technical constraints. Short-range wireless technologies such as Wi-Fi and Bluetooth offer limited coverage and typically incur relatively high power consumption, making them unsuitable for large-scale rural deployments. While cellular technologies such as LTE and 5G can deliver wide-area coverage, their deployment and operational costs, together with energy requirements, often limit their practicality in sparsely populated regions. In this context, Low-Power Wide-Area Network (LPWAN) technologies have emerged as promising enablers for rural IoT, owing to their long communication range, low power consumption, and reduced infrastructure cost. Among these technologies, LoRa has gained particular attention due to its robustness, scalability, and affordability. As highlighted in [51], however, deploying LoRa networks in real rural environments introduces a set of unique challenges related to scalability, reliability, and practical network operation. For instance, Precision agriculture in regions without reliable internet connectivity presents significant challenges, particularly in monitoring and managing irrigation systems. In [52], the authors develop a LoRa-based IoT solution aimed at improving water management in connectivity-constrained agricultural environments. A key motivation of the work is the difficulty of detecting leaks and ensuring sufficient water delivery to crops when irrigation pipelines are distributed over large areas. The proposed system integrates sensing and control nodes that communicate via LoRa to monitor water flow and regulate irrigation through solenoid valves. A gateway node enables intermittent internet access for data visualization and system control, allowing users to remotely identify issues such as leaks or inadequate water supply. By enabling timely detection and corrective action, the system supports sustainable farming practices and demonstrates how LPWAN-enabled IoT solutions can enhance traditional agricultural operations in regions lacking continuous internet connectivity.

7.1.3. Agricultural monitoring based on Internet-of-Things and UAV

As information technology advances, Internet of Things (IoT) systems and low-altitude remote sensing platforms such as unmanned aerial vehicles (UAVs) are increasingly applied in environmental and agricultural monitoring. In agricultural contexts, IoT devices enable the collection of weather and environmental parameters related to crop growth using low-cost sensors, while UAVs provide aerial imagery of farmlands that can be analyzed to detect the presence of pests or crop diseases. In this regard, Gao et al. [53] propose an agricultural monitoring framework aimed at improving insights into the occurrence of pests and the influence of weather parameters. To address the energy limitations associated with deploying systems in remote farmland, the authors introduce a sun-tracking mechanism to enhance solar energy harvesting, as well as a wind-assisted UAV flight mode to extend flight duration. UAV-acquired images are transmitted to a cloud-based platform for disease and pest impact analysis. Field deployment in the Yangtze River region indicates that wheat is more susceptible to disease at temperatures between 14 °C and 16 °C, while higher rainfall reduces the spread of wheat powdery mildew.

Overall, these deployment examples demonstrate the practical feasibility of NTN-assisted IoT systems, particularly those incorporating satellites and UAVs, while also underscoring the recurring importance of energy-efficient communication and the prominent use of LPWAN technologies such as NB-IoT and LoRa, as reflected in both the reviewed literature and the underlying bibliometric co-occurrence patterns. At the same time, the scarcity of real-world deployments involving technologies such as NOMA, 5G, cognitive radio, edge intelligence, and RIS highlights the need for further experimental and field-based studies to validate their practical applicability in NTN-assisted IoT scenarios.

8. Discussion and research outlook

The thematic clusters derived from the bibliometric analysis offer a structured lens through which to understand the depth of current research in ML-based NTN-assisted IoT, as well as implications for future research. In this section, these clusters were further discussed to provide an understanding of the intellectual contributions and a foundation for identifying and discussing opportunities, gaps, and synergies between research streams. This can provide further insights to align the existing research contributions with technological advancements (e.g., IoT, image processing, mobile edge computing, autonomous navigation, etc.) and societal needs. In particular, the discussion is anchored in insights from the identified clusters, which serve as the primary lens for interpreting the roles of enabling technologies, practical deployment constraints, and emerging research opportunities in NTN-assisted IoT systems.

8.1. Enabling energy-efficient NTN-IoT through MEC, NOMA, and IRS

Energy-Efficiency Considerations: Energy efficiency is vital in wireless communication applications, and NTN-assisted IoT is no exception. As the world moves toward higher data rates with ultra-high reliability requirements, it is important to ensure that these technologies remain energy efficient. In this section, we discuss several technologies that can support NTN-assisted IoT, especially UAV-assisted IoT, and examine some of the benefits and challenges they introduce. Some of these technologies extend beyond IoT applications, and special focus is given to NOMA, cognitive radio, and intelligent reflecting surfaces, which are especially prominent due to their high number of connections within this research landscape.

UAV-Edge Integration: The paradigm of B5G wireless communication smart applications requires the deployment of intelligent technologies. For example, UAVs can support high-computation tasks as MEC nodes, allowing better storage and processing of massive data. This is especially important because IoT nodes have energy and computational limitations that restrict their capacity to handle traditional AI algorithms for extracting useful information from data-intensive tasks. Deploying UAVs as MEC nodes can improve the quality of service for these applications. Moreover, AI algorithms such as federated learning have the potential to optimize network performance across multiple metrics, which include task offloading, latency, energy efficiency, and security vital in IoT-based smart applications [54].

Spectral Efficiency via NOMA: NOMA is one of the most promising technologies to help 5G and future-generation networks reach their spectral efficiency targets. The deployment of complementary techniques, such as massive multiple-input multiple-output (MIMO) and blockchain, can further enhance these networks. Combining NOMA with these technologies can significantly improve overall system performance, especially in 5G cooperative cognitive radio networks where NOMA power-domain schemes are used to enhance downlink spectral efficiency. For example, [55] investigates a novel cognitive NOMA-based cooperative CRN for underlay spectrum sharing. In this work, cooperative NOMA is proposed by carefully considering the access modes of secondary users and relays, and its efficiency is evaluated under realistic operating conditions.

Task Allocation Optimization: The process by which tasks are distributed among various nodes in an edge computing network is called task allocation. However, determining the optimal location for each task, considering requirements such as processing power, storage, and network bandwidth, while also adapting to the highly dynamic network conditions, poses a significant optimization challenge. Several approaches have been proposed to address this problem, and they generally fall into categories spanning from classical centralized and decentralized frameworks to modern learning-based solutions. Selecting the most suitable task allocation strategy depends on the specific edge computing architecture and its deployment model, for instance, MEC, cloud–edge, fog computing, or peer-to-peer edge computing. In particular, task allocation in MEC networks can be especially complex due to the need to balance multi-objective trade-offs, including energy efficiency, security, data privacy, quality of service (QoS), and latency. Properly comparing these methods is therefore vital [56].

CR-NOMA Hybridization: One promising solution to meeting the requirements of beyond-5G networks, especially for IoT, is the integration of cognitive radio and NOMA technologies. This integration offers many benefits. For instance, power-domain NOMA enables multiple users to share the same orthogonal resource blocks, while cognitive radio enables opportunistic access to licensed frequency bands by allowing secondary users to utilize spectrum without disrupting primary users' activities. To support massive IoT connectivity requirements, a number of multiple-access techniques have recently been combined with CR-based NOMA systems. Moreover, the deployment of emerging technologies such as unmanned aerial vehicles, intelligent reflecting surfaces, machine learning, hybrid NOMA, and terahertz communications holds great promise for addressing various challenges in CR-NOMA networks [57].

Joint Resource Optimization: Multiple IRS- and UAV-assisted MEC systems offer significant benefits for 5G/6G networks. Deploying multiple UAVs with multiple IRSs can efficiently accommodate large-scale deployments of user equipments (UEs). However, energy consumption, long mission completion times, and UAV maintenance costs pose major challenges. Addressing these challenges requires joint optimization of UAV trajectories, IRS phase shifts, stopping point locations, and order, as well as the association between UEs and UAVs at each stopping point [58]. Solving this problem effectively requires the deployment of efficient algorithms, such as artificial intelligence-based solutions, to ensure efficient network operation.

Trajectory and Power Control: UAV-enabled MEC in IoT systems offers a promising solution for achieving low latency and high energy efficiency in next-generation (5G/B5G) IoT-enabled applications. Given the limited spectrum availability, novel technologies are required to address this constraint, and NOMA provides a spectrally and power-efficient communication solution, especially for task offloading. For

example, in [59], the authors aim to maximize the sum of bits offloaded from IoT devices by jointly optimizing the IoT device–UAV association, the transmit power of IoT devices, the energy budgets of both IoT devices and UAVs, as well as the UAV trajectory. Solving such optimization problems efficiently can be achieved using advanced machine learning techniques such as deep reinforcement learning.

Disaster Recovery and Resiliency: NTNs find important applications in disaster relief to support user connectivity. In such scenarios, it is vital to optimize computing capacity, radio resources, and the UAVs' trajectories at each time slot in order to maximize the number of satisfactory connections. UAV-based aerial base stations (ABSs) can help establish links for users using power-domain NOMA. Given the potential inoperability of terrestrial networks during disasters, meeting QoS requirements, especially end-to-end (E2E) delay, and ensuring satisfactory data transmission for UEs is critical. Artificial intelligence techniques, such as multi-agent deterministic policy gradient (DPG) deployed in [60], can be very effective. Specifically, this algorithm enables ABSs to collaborate and maximize the number of satisfactory connections by interacting with the environment and learning an optimal policy.

Interference Management: The implementation of the NOMA protocol enables multiple-access capabilities for UEs by managing multiple access interference in ultra-dense 5G edge cloud environments. These networks typically consist of macro and micro base stations along with different types of UEs. However, efficient task scheduling and resource allocation are required for these applications, and the dynamic nature of user tasks makes scheduling even more challenging. In this regard, efficient AI-based algorithms can help optimize task scheduling and resource allocation while minimizing transmission energy consumption. In particular, the goal is to find an optimal schedule that maximizes response rates, considering challenges such as user mobility, workload variations, request sizes, and the number of mobile users [61].

8.2. Emerging intelligent technologies for 6G NTN and IoT applications

6G Service and its enablers: Although 6G promises to offer several enhanced capabilities suitable for diverse IoT applications, the integration of machine learning into 6G holds transformative potential to solve the emerging challenges of 6G wireless communication. In particular, this can pave the way for supporting Ultra-Reliable Low-Latency Communications (URLLC), enhanced Mobile Broadband Communications (eMBB), and massive Machine-Type Communications (mMTC) with greater capabilities compared to 5G. Thus, facilitating applications such as augmented reality, virtual reality, and ultimately extended reality (XR) [62].

RIS-Enhanced Architectures and Role of AI: Integrating RIS has transformative potential in 6G NTNs [63]. RIS can address diverse user requirements and can accommodate components such as MIMO and advanced radio communication. These requirements include higher data rates, power efficiency, secure data transmission, and extended coverage. In such architectural setups, AI and ML play a vital role in network optimization. For instance, using RIS-based beamforming to ensure sustainable and energy-efficient NTN operations [64].

Dynamic Network Slicing: It is expected that 5G, as a successor to 4G, will create new, dynamic, differentiated, and transparent services. The arrival of 5G brings demand surges along with new expenses and use cases. However, since these use cases are diverse and require more complex network architectures, resource management and service orchestration becomes increasingly complex. To address this challenge, one potential technology is network slicing, which enables efficient resource allocation and supports self-service capabilities. Nevertheless, network slicing has its own inherent challenges. To implement efficient network segmentation, it is important to develop robust algorithms that guarantee optimal resource allocation. Artificial intelligence and machine learning can play a vital role in analyzing huge datasets to facilitate intelligent decision-making processes; however, the acquisition of such high-quality datasets remains a significant practical challenge. To overcome this, it is important to develop robust ML solutions to address resource allocation and dynamic network slicing orchestration requirements to improve the efficiency of next-generation mobile networks [65].

NOMA for Global Connectivity: NTNs have the potential to support global connectivity for 6G ubiquitous wireless systems. One of the most promising technical candidates for next-generation 6G communication is non-orthogonal multiple access (NOMA), as it can facilitate massive connectivity while improving spectral efficiency. The synergy between NTN and power-domain NOMA is integral to addressing several problems related to 6G. NOMA and NGMA have the potential to address specific challenges associated with NTNs for 6G. When combined with emerging technologies such as integrated sensing and communication, mmWave, and THz, they can significantly improve spectral efficiency and enhance urban air mobility and IoT applications for increased connectivity [66]. Optimizing NTN NOMA requires efficient AI algorithms as well as mathematical optimization and learning tools to tackle its inherent challenges.

Efficient Resource Management: 5G aims to serve different market verticals by providing wireless connectivity that meets diverse QoS requirements. Efficient resource management is required to meet the demands of these verticals while ensuring optimal utilization of network resources at the core, transmission, and radio access layers. Network slicing is one of the key 5G technologies that enables this by creating virtualized subnetworks that isolate resources based on the unique requirements of each vertical. To effectively address the stochastic nature of these demands, Markov decision process frameworks can be deployed to model radio resource management in 5G radio access network slices. In such cases, inter-slice radio resources can be efficiently allocated to support highly reliable and bandwidth-hungry applications [67]. In general, schedulers developed to solve such problems should take into account network load and channel conditions for efficient resource allocation. Similarly, proposed schedulers must provide an optimal allocation strategy for sharing bandwidth between URLLC slices and eMBB slices.

Aerial-Satellite Channel Dynamics: NTNs have huge potential to support terrestrial IoT applications, especially for smart cities with diverse use cases. In particular, satellite-based IoT and UAV-based IoT are promising technologies. However, challenges such as the dynamic nature of satellite and UAV channels, as well as the high maneuverability of UAVs, make them more complex to implement. Addressing these issues requires solutions at both the physical and non-physical layers, as well as effective route planning for UAVs [68].

Space-IoT Integration: Recent developments in IoT, machine learning, AI, distributed computing, and data communication technologies have led to significant breakthroughs and created opportunities across various domains. These range from fast, reliable, and efficient data communication to large-scale computing at the cloud or edge, as well as intelligent data analytics. Importantly, these technological advancements have also opened up new possibilities in space and non-terrestrial environments. In particular, the integration of IoT with space-based technologies has enabled numerous promising applications [69].

IoT Research Frontiers: There are numerous active research areas related to IoT technologies. These include the development of future wireless communication systems, such as 5G and 6G, which require reliable machine-to-machine (M2M) communication. Other areas of focus involve engineering the future Internet, designing IoT hardware to support both low-power and long-range data transmission and processing, and advancing localization technologies for humans, robots, and devices, often through satellite-based methods. Research is also being conducted on middleware and platforms to support IoT applications, as well as on IoT standards and semantic technologies. Additionally, improvements in IoT infrastructure, including sensors and actuators, along with associated data management and processing techniques,

are critical. Finally, addressing key socio-technical challenges such as security, privacy, and ethics remains an essential area of ongoing research [69].

Mission-Critical Service Expectations: In recent years, IoT applications have become pervasive across many aspects of daily life, including scenarios that are highly critical and demand strict reliability. These mission-critical applications, such as disaster management, military operations, and rescue efforts, rely on IoT technologies to deliver both sustainability and accuracy. IoT is uniquely suited for situations where human intervention is limited or unsafe due to life-threatening conditions. In such environments, IoT sensors can gather vital data and transmit it to processing hubs for timely decision-making. The integration of IoT with satellite networks has the potential to significantly enhance service delivery in these contexts by expanding coverage and improving reliability. For example, the hybridization of Narrowband IoT (NB-IoT) with satellite networks can support long-term sustainability, particularly due to NB-IoT's energy-efficient characteristics [70].

Security and SDN Defense: The massive increase in IoT deployments, along with the integration of AI and ML technologies, has paved the way for automation across a range of applications, including manufacturing, automobiles, healthcare, agriculture, and communication. AI-driven IoT will also play a crucial role in space exploration technologies, where IoT networks will benefit from broad coverage, largely enabled by the integration of low Earth orbit satellites. However, the adoption and expansion of these technologies depend heavily on ensuring data security and preventing malicious intrusions. Achieving this will require the use of advanced machine learning and AI algorithms to safeguard against data breaches in such applications. Also, SDN-based traffic monitoring and regulation systems, when well implemented, can help identify and block malicious traffic [71].

Cyber-Physical Systems Optimization: Processing data from Internet of Things (IoT) devices and cyber–physical systems plays a critical role in optimizing performance within pervasive computing environments. In particular, evaluating the performance of applications such as autonomous robotics and motion capture systems requires data-driven techniques, environmental parameter analysis, AI algorithms (such as decision trees), and data mining. Techniques like machine data mining, cognitive data visualization, predictive modeling, and real-time data processing contribute significantly to enhancing the performance of industrial robots and the quality of their outputs. Optimization of industrial robotic systems also relies on unmanned robotic networks, specialized operating systems, product simulation models, spatio-temporal fusion algorithms, and decision support tools. Similarly, predictive simulation tools, computer vision algorithms, process mining techniques, digital twin technologies, cloud computing, and data analytics all play vital roles. These technologies enable data-driven decision support and improve performance in robotic manufacturing environments [72].

IIoT Connectivity Expansion: During the last few years, various Industrial Internet of Things (IIoT) applications have emerged with numerous network elements interconnected using wired and wireless communication technologies and equipped with strategically placed sensors and actuators. This paper [73] justifies why NTNs will bring the IIoT vision closer to reality by providing improved data acquisition and massive connectivity to sensor fields in large and remote areas. NTNs encompass a multi-layered infrastructure that integrates satellites, HAPs, and aircraft to extend radio coverage and provide remote monitoring and sensing services. Additionally, this paper describes indicative delay-tolerant massive IIoT and delay-sensitive mission-critical IIoT applications spanning a large number of vertical markets with diverse and stringent requirements. As the heterogeneous nature of NTNs and the complex and dynamic communications scenarios lead to uncertainty and a high degree of variability, conventional wireless communication technologies cannot sufficiently support ultra-reliable and low-latency communications (URLLC) and offer ubiquitous and uninterrupted interconnectivity. In this regard, [73] sheds light on the potential role of artificial intelligence (AI) techniques, including ML and deep learning (DL), in the provision of challenging NTN-based IIoT services and provides a thorough review of the relevant research works. By adding intelligence and facilitating the decision-making and prediction procedures, the NTNs can effectively adapt to their surrounding environment, thus enhancing the performance of various metrics with significantly lower complexity compared to typical optimization methods. IIoT applications span both delay-tolerant and delay-sensitive mission-critical use cases, covering a wide range of vertical markets with diverse and stringent QoS requirements. These applications involve numerous interconnected network elements utilizing both wired and wireless communication technologies, supported by strategically placed sensors and actuators. The introduction of NTNs aims to enhance IIoT by enabling improved data acquisition and connectivity in remote and large geographical areas. In particular, NTNs employ satellites and aerial vehicles to extend radio coverage and provide remote sensing and monitoring services. However, the integration of NTNs introduces inherent challenges due to their heterogeneous nature and the complex, dynamic communication scenarios they present, resulting in a high degree of uncertainty and variability. To address these challenges, AI, including ML and deep learning (DL), holds promise for enhancing decision-making, predictive mechanisms, and environmental adaptation. These techniques can significantly improve performance metrics and reduce system complexity.

9. Future directions

Building on the dominant themes identified through the initial bibliometric analysis, this section explores emerging research trends by isolating low-frequency keywords (with fewer than five occurrences). While these terms did not meet the primary threshold for the main clustering results, a refined secondary cluster analysis was performed on this subset to capture high-granularity research directions often obscured by higher keyword occurrence thresholds. From this analysis, six thematic groups were strategically selected based on their relevance to the paper's scope (see Table 6). For each subsection, the representative keywords from these clusters are first presented to define the thematic boundaries, followed by an in-depth discussion of their implications for deployable NTN-assisted IoT systems

9.1. Physical-layer challenges and service innovations for next-generation IoT

Keywords: adaptation, agriculture drones, complexity, decoding, eMBB, encoding, flying sensor networks, geospatial analytics, innovations, internet of things (IoT), IoV, Ku-band, low earth orbit (LEO), low earth orbit (LEO) satellite, machine-type communication, millimeter-wave band, mMTC, mobile communication, modulation, NewSpace, physical layer, smart vehicles, URLLC.

ML is expected to play a crucial role in enabling non-terrestrial network (NTN)-assisted IoT applications, paving the way for innovative services such as smart vehicles, agriculture drones, flying sensor networks, and large-scale machine-type communications (mMTC) with massive numbers of connected devices. These applications, powered by satellite technologies (e.g., LEO satellites) and commercial space ventures (e.g., NewSpace), require significant optimization at the physical layer to ensure reliability, scalability, and efficiency. In line with the application-driven emphasis on energy efficiency and next-generation wireless communication, these physical-layer innovations are particularly relevant where energy efficiency, reliability, and large-scale access mechanisms directly shape deployable 5G/B5G/6G NTN-assisted IoT services. A key challenge is that NTN physical-layer conditions, particularly in UAV-assisted links, are often highly non-stationary due to mobility, blockage, and unstable propagation environments. Satellite communications also exhibit distinct propagation impairments arising

Table 6
Clusters and their associated keywords (including those with fewer than five occurrences).

SN	Associated keywords
1	adaptation, agriculture drones, complexity, decoding, eMBB, encoding, flying sensor networks, geospatial analytics, innovations, internet of things (IoT), IoV, Ku-band, low earth orbit (LEO), low earth orbit (LEO) satellite, machine-type communication, millimeter-wave band, mMTC, mobile communication, modulation, NewSpace, physical layer, smart vehicles, URLLC
2	4G, ConvLSTM, generative AI, industrial IoT, internet of everything, long short-term memory, mobile networks, network intrusion detection system, O-RAN, optical wireless communication, precision livestock management, search and rescue, semantics, sensing, spoofing, tactile internet, technologies, terahertz, THz communications, tracking, V2X
3	6G communications, backscatter communication, caching, communications, connectivity, deep Q-learning, energy-efficient, graphene, MDP, MIMO antenna, networking, SAGIN, sensor data, spectrum access, temperature, wildfire, wireless
4	5G and B5G communication, channel prediction, hybrid beamforming, massive MIMO, mmWave communication, network virtualization, QUIC, road safety, software-defined network, virtualization
5	cognitive radio network, cyberattacks, energy management, intelligent transportation, internet of flying things, network sensing, privacy preservation, software-defined radio
6	artificial neural network, cognitive radio, opportunistic spectrum access (OSA), signal detection, spectrum sensing, structural health monitoring (SHM), trajectory control

from long distances and atmospheric conditions. These dynamics can significantly degrade channel estimation accuracy, motivating the use of learning-based estimation techniques. If such channel uncertainties are not explicitly accounted for, learning-driven optimization of higher-level policies (e.g., trajectory or altitude control) may become inefficient or unreliable.

Future research directions include the design of efficient encoding and decoding schemes, as well as the development of novel modulation techniques tailored for high-mobility and high-density IoT scenarios. The exploitation of millimeter-wave bands offers opportunities for enhanced mobile communications, though challenges such as link stability and environmental susceptibility must be addressed. Moreover, the creation of low-complexity algorithms will be vital to maintain energy efficiency and reduce computational overhead in constrained IoT devices. In addition to algorithmic design, realistic evaluation under practical considerations such as mobility and imperfect CSI is essential to ensure that proposed physical-layer innovations truly leads to robust performance gains in real-world NTN-assisted IoT deployments. Finally, leveraging spectrum resources such as the Ku-band for satellite-assisted IoT systems represents a promising avenue for delivering ultra-reliable, low-latency, and massive connectivity in future IoT ecosystems.

In addition, AI-driven solutions such as reinforcement learning (RL) show strong potential for adaptive modulation, where they can dynamically select modulation schemes by learning from SNR and CSI conditions. This approach has the potential to improve bit error rates and throughput under varying SNR levels, thereby enhancing Quality of Service (QoS) compared to fixed modulation schemes [74]. ML algorithms can also address outdated channel quality indicator (CQI) problems that arise when feedback delays hinder timely adjustments of downlink modulation order and coding rate. In such cases, ML can be employed to estimate updated SINR values, which can then be translated into CQI levels in 5G and B5G networks adopting adaptive modulation [75].

RL-based frameworks are also promising for optimizing beamforming vectors and power control in massive MIMO systems. Methods such as Deep Q-Networks (DQN) and Distributional Soft Actor-Critic with Three Refinements (DSAC-T) have been demonstrated to improve spectral efficiency, energy efficiency, and latency performance, while mitigating policy overestimation and instability in dynamic channel conditions [76,77]. These advances illustrate the potential of deep reinforcement learning to provide real-time beamforming and power allocation, though challenges of scalability, parameter selection, and computational complexity remain important research considerations.

Similarly, deep learning has been applied to enhance channel estimation by exploiting spatial–temporal correlations in wireless environments [78,79]. Meanwhile, federated learning offers a distributed paradigm that is particularly relevant for large-scale IoT deployments, as it enables collaborative model training across devices without requiring raw data exchange. This approach can tackle privacy, scalability, and communication efficiency challenges, making it an attractive solution for future-generation IoT networks. Ultimately, future research should focus on developing scalable AI solutions that sustain performance gains in the presence of channel uncertainty, user mobility, and environmental variability.

Beyond improvements at the physical layer, application-driven solutions are required that combine high–data-rate communication technologies with advanced analytics, including those derived from UAV-acquired data. For instance, NTN-supported agricultural applications can leverage drone-based geospatial analytics, while smart city and vehicular systems can exploit the ultra-high data rates enabled by beyond-5G networks. These use cases call for continued innovation in modulation schemes, millimeter-wave integration, and adaptive learning-based techniques to address persistent physical-layer challenges. This, in turn, underscores the importance of improving physical-layer adaptivity, such as dynamic modulation and beamforming, to better align communication performance with application-level requirements, including timeliness and reliability in IoT services.

9.2. AI-enhanced communication and emerging technologies for IoT and beyond

Relevant Keywords: 4G, ConvLSTM, generative AI, industrial IoT, internet of everything, long short-term memory (LSTM), mobile networks, network intrusion detection system, O-RAN, optical wireless communication, precision livestock management, search and rescue, semantics, sensing, spoofing, tactile internet, technologies, terahertz (THz), THz communications, tracking, V2X.

Communication technologies integrated with NTN-assisted IoT and advanced machine learning algorithms are expected to form the foundation for next-generation applications such as the tactile internet, internet of everything (IoE), industrial IoT, and mission-critical use cases, including search and rescue, precision livestock management, vehicular communications (V2X), and network intrusion detection systems. These technologies must be optimized to ensure seamless convergence of communication, sensing, and semantic information exchange. In this context, generative AI offers promising capabilities to enhance semantic communication, synthesize realistic data for training, and support adaptive decision-making in highly dynamic network environments. A key open issue is the need for rigorous benchmarking and reproducibility of generative AI methods, as well as mechanisms to mitigate bias and hallucination, which pose significant risks in safety-critical NTN-assisted IoT applications.

Also, future research should explore the design of efficient tracking algorithms and the application of emerging communication paradigms such as terahertz (THz) communications, optical wireless communication, and open radio access networks (O-RAN). ConvLSTM, which extends fully connected LSTM with convolutional structures in both

input-to-state and state-to-state transitions, enables simultaneous modeling of spatiotemporal correlations and has been widely applied in predictive tasks such as weather forecasting [80]. This capability can be effectively leveraged for predictive IoT applications, including traffic forecasting and mobility-aware resource management. Similarly, GAN-based methods have demonstrated strong potential for anomaly detection [81], making them valuable tools for intrusion prevention in IoT systems [82]. Reinforcement learning-based approaches are also essential for adaptive spectrum management, ensuring optimized spectrum utilization while balancing system-wide performance objectives. Finally, semantic communication is an emerging research direction in which AI can play a crucial role in ensuring that the most relevant and inferred information is communicated in a resource-efficient manner, while simultaneously addressing security vulnerabilities and attack resilience. These advancements demonstrate that AI solutions can address challenges across multiple network layers in NTN-assisted IoT systems while simultaneously enhancing network security, and therefore merit focused exploration in future research. In addition to data-driven communication and semantic processing, ensuring robustness and security in intelligent reconfigurable environments remains an important research direction. Recent works such as [83,84] have studied physical-layer security in satellite-terrestrial and space–air–ground integrated networks, illustrating how RIS-assisted transmission and endogenous wireless security mechanisms can enhance secure communication in heterogeneous NTN environments. Accordingly, future research on NTN-assisted IoT should jointly consider semantic inference, scalable channel access mechanisms, and reliability enhancement, particularly for large-scale deployments involving UAV- and satellite-based connectivity.

Overall, advancements in machine learning have the potential to transform next-generation IoT services, ranging from real-time monitoring and control in industrial IoT, to autonomous decision-making in UAV-IoT communications, precision livestock management for sustainable agriculture, and resilient search-and-rescue systems. Furthermore, future-generation integrated sensing and communication (ISAC) applications can also benefit significantly from these AI-driven innovations, enabling seamless convergence of communication, sensing, and computation.

9.3. AI-driven spectrum and energy optimization in 6G/NTN IoT

Keywords: 6G communications, backscatter communication, caching, communications, connectivity, deep Q-learning, energy-efficient, graphene, MDP, MIMO antenna, networking, SAGIN, sensor data, spectrum access, temperature, wildfire, wireless.

State-of-the-art hardware and spectrally efficient access technologies, such as graphene-based transceivers and massive MIMO antennas, are critical to supporting the extremely high data rates required by 6G communications and 6G-enabled IoT applications. These capabilities are particularly vital in emergency scenarios that demand high-resolution video streaming and multimodal sensing, such as wildfire detection, where real-time visual and temperature data transmission is essential. To ensure sustainable deployment, solutions must also address caching and data storage strategies while meeting stringent connectivity requirements. Energy efficiency remains a fundamental prerequisite for guaranteeing long-term service delivery, especially for battery-constrained IoT devices operating in mission-critical conditions. These requirements motivate the need to adopt advanced communication and computing technologies that jointly address spectral efficiency, high data rates, caching, and energy consumption, as well as learning-based optimization methods capable of managing dynamic trade-offs under time-varying channel and propagation conditions.

In this context, the integration of satellite, aerial, and terrestrial segments within space–air–ground integrated networks (SAGIN) provides an effective framework to extend global IoT connectivity. Reinforcement learning methods have shown great promise in optimizing dynamic spectrum access and energy allocation, particularly in environments with partially predictable patterns. AI frameworks based on Markov decision process (MDP) formulations can facilitate optimization through deep reinforcement learning algorithms such as Deep Q-Learning, as well as value-based, policy-based, and actor–critic methods. These approaches can optimize system objectives while balancing key performance metrics such as throughput, latency, and energy consumption in heterogeneous 6G-enabled NTN-assisted IoT deployments. Furthermore, backscatter communication, which utilizes ambient signals for passive transmission [85], offers a promising solution for low-power connectivity in UAV-assisted applications and large-scale sensor/IoT networks, thereby supporting energy-efficient network deployments. A key research gap lies in developing scalable learning frameworks that remain effective under partial observability, particularly in large-scale NTN-assisted IoT deployments involving multiple agents, such as multi-UAV systems or integrated satellite–UAV networks.

Future research should address cross-layer optimization challenges in NTN-assisted IoT by applying state-of-the-art machine learning and reinforcement learning algorithms. These approaches should jointly consider spectrum access, energy-efficient solutions such as energy harvesting and backscatter communication, and system resilience for disaster-response applications, including wildfire detection and monitoring. Such solutions can be tightly integrated with NTN technologies, particularly satellites, which are well-suited to provide coverage and communication in scenarios where sensors are deployed for situational awareness.

AI-driven methods in NTN-assisted IoT can also support weather forecasting and climate monitoring through distributed sensors that measure parameters such as temperature and humidity. Moreover, they can aid in post-disaster recovery by restoring connectivity and transferring critical information, such as the locations of affected individuals. Beyond disaster contexts, these solutions can also provide high-speed and consistent connectivity in urban deployments, such as smart cities. Collectively, such applications highlight the potential of spectrum-efficient, AI-enabled solutions to deliver valuable and sustainable services in beyond-5G and NTN-supported IoT ecosystems.

9.4. Bridging physical and network layer innovations in B5G for mission-critical IoT

Keywords: 5G and B5G communication, channel prediction, hybrid beamforming, massive MIMO, mmWave communication, network virtualization, QUIC, road safety, software-defined network (SDN), virtualization.

ML is expected to play a transformative role in bridging physical- and network-layer innovations within 5G and beyond-5G (B5G) communication systems, particularly for NTN-assisted IoT applications. At the physical layer, ML algorithms have been successfully applied to channel prediction and estimation [63], dynamic beamforming optimization [86], and channel state information (CSI) recovery [86]. Such techniques are crucial for addressing the high propagation and penetration losses associated with mmWave communications [87]. In particular, hybrid beamforming and intelligent beam management are central to achieving the required Quality of Service (QoS) in massive MIMO systems [88]. Advanced deep learning methods, such as 3D convolutional neural networks for beam selection in V2X applications, can also reduce power dissipation while enhancing vehicular communication reliability [89].

At the network layer, AI can support traffic management and resource optimization in 5G by enabling intelligent congestion control [90], dynamic spectrum management, and autonomous orchestration of resources across diverse service demands [91]. These capabilities are especially critical in network slicing, where multiple virtualized networks coexist on a shared physical infrastructure. AI-enabled orchestration frameworks, including federated learning approaches, can enhance scalability and adaptability to real-time service

demands [92]. Furthermore, reinforcement learning and adaptive optimization techniques can be applied to bandwidth allocation and slice selection, ensuring both energy efficiency and robust QoS under varying workloads [93]. For mission-critical IoT, an important direction is to ensure that AI-driven orchestration can deliver predictable reliability and support security requirements under dynamic network conditions and heterogeneous NTN environments involving UAVs, satellites, and high-altitude platforms.

In parallel, AI can improve the performance of emerging protocols such as QUIC by enabling intelligent congestion control, latency reduction, and improved reliability for mission-critical services. Software-defined networking (SDN) and virtualization further provide programmable environments for adaptive network management, though challenges remain in terms of security and privacy risks [94]. Addressing these vulnerabilities requires integrating intelligent AI-based solutions. For instance, AI can support efficient anomaly detection [95], intrusion prevention [96], and other privacy-preserving solutions for mission-critical IoT services such as those involving intelligent transport applications. For instance, reliable V2X communication is essential for road safety in autonomous driving applications. Machine learning can solve diverse problems in vehicular networks. For instance, machine learning algorithms can be deployed to solve security challenges such as gray hole attacks, Distributed Denial of Service detection, trust computation, Sybil attacks, fingerprinting attacks, spoofing attacks, and jamming attacks [97].

Overall, exploiting AI to jointly optimize physical-layer aspects (channel estimation, beamforming, prediction) and network-layer solutions such as (network slicing, virtualization, protocol optimization) is essential for supporting latency-sensitive, ultra-reliable applications such as V2X, road safety, and emergency response. Future research should prioritize the utilization of learning algorithms to support diverse IoT applications toward seamless connectivity while respecting overall system constraints. In addition, bibliometric co-occurrence patterns indicate that energy efficiency and spectrum-access mechanisms are likely to remain central design considerations in NTN-assisted IoT research, reinforcing the need to study their joint optimization through learning-based strategies.

9.5. Integrating cognitive radio, SDR, and security for next-generation IoT applications

Keywords: cognitive radio network, cyberattacks, energy management, intelligent transportation, internet of flying things, network sensing, privacy preservation, software-defined radio (SDR), artificial neural network, opportunistic spectrum access (OSA), signal detection, spectrum sensing, structural health monitoring (SHM), trajectory control.

Future applications of ML will increasingly involve autonomous aerial platforms, where UAVs perform trajectory control to support structural health monitoring (SHM). In such cases, drones, forming part of the emerging *internet of flying things*, can detect cracks or defects in infrastructure and communicate this information to ground or aerial base stations, thereby preventing catastrophic failures and loss of life. Privacy preservation remains a critical challenge in these scenarios, since sensitive monitoring data may be exposed to interception or manipulation. Additionally, UAVs and flying sensors may share spectrum resources with traditional cellular users, particularly in urban environments, creating an urgent need for intelligent spectrum management. Efficient trajectory control is therefore a key requirement in critical aerial sensing applications such as structural health monitoring, where learning-based approaches can assist in adapting flight paths to balance energy consumption and communication reliability under diverse air-to-ground sensing and propagation conditions.

Cognitive radio and software-defined radio (SDR) technologies are expected to significantly enhance spectrum efficiency in 5G and beyond-5G networks supporting NTN-assisted IoT. ML algorithms can predict spectrum usage patterns and prioritize access to spectral resources under dynamic conditions. Opportunistic spectrum access (OSA) can be enabled through traditional ML approaches that analyze datasets for temporal and spatial patterns, or via reinforcement learning algorithms that optimize access strategies using adaptive reward functions. Artificial neural networks can also support spectrum sensing [98], network sensing [85], and signal detection [99,100], supporting efficient 5G and B5G for NTN-assisted IoT applications. These approaches improve not only spectral utilization but also energy efficiency, ensuring throughput and reliability in applications such as intelligent transportation systems and heterogeneous IoT networks. An important research direction is to improve the robustness and reliability of cognitive sensing through learning-based mechanisms, particularly in dense and heterogeneous IoT environments where non-cellular devices coexist with traditional cellular systems under congested spectrum conditions.

Security and privacy preservation will be equally important in these deployments. AI-driven anomaly detection frameworks can improve cybersecurity in IoT-enabled network applications operating over 5G, such as smart cities [101]. In 5G and beyond-5G networks, AI can play a vital role in empowering intelligent, adaptive, and autonomous security management, especially since it can identify hidden patterns from a large set of time-varying and multi-dimensional data toward making accurate decisions. However, its vulnerabilities can also be exploited to jeopardize the security of future networks [102], underscoring the importance of future research into explainable, robust, and privacy-preserving AI security mechanisms. These considerations highlight the importance of security-aware and explainable AI approaches, particularly for intrusion detection and resilience against adversarial attacks. In addition, privacy-preserving learning frameworks are increasingly relevant for enabling collaborative model training in distributed NTN-assisted IoT environments without exposing sensitive data and thus should be leveraged in NTN-assisted IoT.

Overall, integrating cognitive radio, SDR, and AI-driven security mechanisms offers a promising pathway to enable mission-critical IoT applications ranging from SHM and intelligent transportation to autonomous aerial sensing systems. Future research should pursue joint optimization frameworks to support next-generation and NTN-assisted IoT applications, with particular focus on dynamic spectrum access, trajectory control, and privacy preservation. A representative use case is cognitive radio-enabled IoT, where such frameworks can enable both efficiency and security.

10. Conclusion

The rapid convergence of UAVs, IoT, edge computing, and advanced AI is reshaping modern communication and sensing systems. This paper analyzed the thematic landscape of NTN-assisted IoT research using VOSviewer-based cluster analysis, revealing several clusters across 2469 bibliometric records. Among these, two clusters were identified as particularly relevant to wireless technologies supporting learning-driven NTN-assisted IoT.

The cluster centered on energy efficiency highlights the interaction between physical- and network-layer technologies, where channel modeling, spectrum access strategies, and AI-driven optimization intersect. Machine learning and reinforcement learning techniques are increasingly applied to problems such as UAV trajectory control, mobile edge computing, RIS-assisted communications, disaster-response networking, and 5G-enabled access, supporting sustainable large-scale IoT operation in NTN environments. Research within this cluster emphasizes the role of complementary enabling technologies, including reconfigurable intelligent surfaces, non-orthogonal multiple access, and cognitive radio, in mitigating adverse propagation conditions and improving energy efficiency in UAV-assisted and satellite-supported IoT systems. A key observation is that energy efficiency consistently appears as a cross-cutting design consideration across multiple 5G- and beyond-5G enabling technologies.

The cluster focused on next-generation wireless communication reflects the integration of emerging access technologies with IoT standards such as NB-IoT and LoRa for environmental monitoring and remote sensing applications in NTN-assisted IoT systems. The internal and external keyword connections within this cluster indicate that wireless communication technologies like network slicing, B5G, 6G extend well beyond isolated use cases, spanning domains such as security, edge and cloud computing, Industry 5.0, localization, cybersecurity, earth observation, agriculture, smart cities, and sustainability. Collectively, these links illustrate how learning-based methods are being explored across diverse application domains to support energy-constrained IoT devices assisted by aerial and satellite platforms.

Taken together, the findings indicate that application-driven NTN-assisted IoT systems are increasingly shaped by the tight coupling of energy efficiency, spectrum access, and intelligent security mechanisms, as reflected in the strong interconnections observed between the studied clusters. Looking ahead, this interplay highlights the importance of cross-layer, AI-enabled frameworks that jointly address spectrum efficiency, energy optimization, and system robustness. Future NTN-assisted IoT systems will therefore depend on embedding intelligence across both physical and network layers to enable resilient and adaptive communication for applications ranging from disaster response and precision agriculture to intelligent transportation and smart cities. Overall, the cluster-driven synthesis presented in this paper provides a consolidated reference for researchers and practitioners seeking to understand dominant research trends and emerging directions that are most relevant to practical NTN-assisted IoT deployments.

CRediT authorship contribution statement

Oluwatosin Ahmed Amodu: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization. **Zurina Mohd Hanapi:** Supervision, Resources, Funding acquisition. **Raja Azlina Raja Mahmood:** Writing – review & editing, Writing – original draft, Investigation. **Chedia Jarray:** Writing – original draft, Investigation. **Faten A. Saif:** Writing – original draft, Investigation. **Huda Althumali:** Writing – original draft, Investigation. **Umar Ali Bukar:** Writing – review & editing, Visualization. **Mohammed Sani Adam:** Writing – review & editing, Visualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the authors utilized ChatGPT and Google Gemini to edit and improve the readability of the draft. Google Gemini was also employed to extract keywords from the VOSviewer items tab. After using this tool/service, the author strictly reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that there is no conflict of interest in this paper.

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